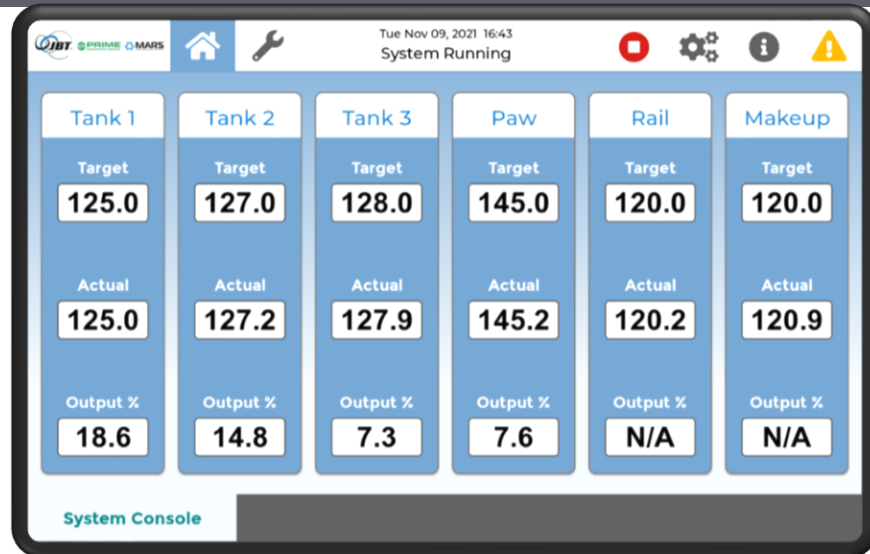


PRIME EQUIPMENT GROUP

JBT SCALDER SYSTEM



Customer: Wayne Sanderson
Union Springs, AL, USA

Sales Order #: 209734S

Serial #: SN4448

Safety Instructions & Technical Manual

Prepared by: Lisa Stephens
Engineer Approval: Justen Vrabel
Approved by: Ashley Walter

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Introduction

We at JBT/Prime Equipment Group, welcome you as an owner/user of JBT/Prime Equipment Group machines.

JBT/Prime Equipment Group designs and manufactures machines that can be operated and maintained safely; however, ultimate personal safety when operating and/or servicing a machine is a shared responsibility between the machine purchaser, and all personnel who operate and service the machine. This shared responsibility, along with other critical safety information, is explained in this section of the manual.

Every effort has been made to provide information and documentation that is representative of the purchased machine model. Serious injury, including death can occur if a machine is not operated and/or serviced in a safe manner. Contact our Customer Care Department if you are not sure of the safe way to perform a task or have questions which could affect the safety of you and your co-workers.

Responsibilities

Machine Builder's Responsibility

JBT/Prime Equipment Group's primary goal is to design machines in a manner that eliminates or minimizes potential hazards.

When potential hazards cannot be designed out or safeguarded, it is JBT/Prime Equipment Group's responsibility to alert machine users of the residual risk(s). JBT/Prime Equipment Group does this using safety signs mounted near or within the hazardous area. JBT/Prime Equipment Group also uses Caution, Warning, and Danger statements throughout the Instruction manual(s) to alert machine users of potential hazards and instruct them on measures to take to avoid personal injury.

We are confident that our machines meet OSHA and other worldwide safety codes and standards.

Machine Purchaser's Responsibility

According to the OSHA act of 1970, each employer (machine purchaser) has the general duty to furnish each employee (machine user) employment and places of employment which are free from recognized hazards that could cause death or serious physical harm. The employer is also required to comply with standards, rules, and regulations proclaimed by the Secretary of Labor and to post notices which inform the employees of their rights and duties as outlined by all applicable laws. The employer is expected to take necessary action, including the establishment and enforcement of rules, to ensure compliance. Most countries employ the same or similar rules on their respective employers.

It is the responsibility of the employer (machine purchaser) to:

- Provide a work environment free from recognizable hazards and promote the expectation of safe work practices.
- Develop, implement, and enforce a comprehensive safety program for the protection of the employees.
- Provide a comprehensive lockout/tagout policy for employees to follow when adjusting/servicing the machine. Provide locks and lockout devices and "Do Not Start" tags to be used when disconnecting/dumping electrical, pneumatic, hydraulic, and mechanical power and stored energy sources as required under the lockout/tagout policy.

NOTE: "Do Not Start" tags CAN NOT be used as the only means of lockout, they must be used in conjunction with approved locks and/or other lockout devices.

- Provide training for employees that will be operating and servicing the machine, on the safe way to operate and service the machine.

NOTE: Your company can purchase a custom, comprehensive training program for machine users from JBT/Prime Equipment Group.

- Prohibit machine operation when any one of the safety devices is not functional (guards, interlocks, lifelines, safety signs, etc.).

Extra copies of the manuals and literature can be purchased through JBT/Prime Equipment Group's Customer Care Department.

GENERAL INFORMATION

SAFETY

Machine User's Responsibility

Machine user (operator and service technician) safety and the safety of all others in the work area are dependent on everyone forming and practicing good safety habits and exercising common sense, good judgment, and reasonable care while operating/servicing the machine.

The machine mounted safety signs and the Danger, Warning, and Caution statements throughout the Instruction manual(s) alert the machine user to potential hazards and instructs her/him on measures to take to avoid personal injury. The signs and statements, along with the guidelines included in this section of the Instruction manual, will help you, the user, understand your role in safe machine operation/servicing.

GENERAL INFORMATION

SAFETY

Before You Operate or Service the Machine for The First Time

- Read the instruction manual or vendor literature relevant to the upcoming task before you operate/adjust/service the machine. If you have questions, get answers before you begin the upcoming task.
- Prior to clearing wraps/jams, mechanically adjusting, or servicing the machine, lockout/tagout the machine following your company's lockout/tagout policy pertaining to that type of task.
- Take a walk around the machine, reading all safety signs on the machine and noting the locations of the Emergency Stop devices.

NOTE: Prior to machine operation/servicing, know the locations of all Emergency Stop buttons/lifelines and the locations of potential hazards.

- **NEVER** defeat or remove any of the safety devices. Each one performs a specific function and all of them are provided for your protection.
- **NEVER** perform a task that you do not have company authorization to perform. Know which machine-related tasks your company has authorized you to perform. Notify the appropriate person to perform all other tasks.
- **NEVER** enter the machine area when you are under the influence of alcohol, drugs, or medications that can make you less alert or affect your judgment.

GENERAL INFORMATION

SAFETY

At the Beginning of Each Shift

- Remove rings, watches, necktie, etc. Bind long hair, roll up long sleeves that do not fit snugly to your wrists, tuck in loose shirt tails, and make sure all other clothing fits closely to your body.
- Review machine operation, performance, and problems with the previous operator/service technician. Make sure all emergency stop features are functional (interlocks, buttons, lifelines, etc.). **NEVER** operate a machine if any of the emergency stop features are not functioning properly. Make sure all problems are satisfactorily resolved prior to machine start.
- Make sure the machine is free of jams. Clear jams following the jam clearing information in the Instruction manual. **NEVER** clear a wrap, jam, or debris from the machine when the machine is in motion.
- Make sure the machine and the machine area are free of tools and debris.
- Make sure all machine guards and access panels are installed securely.
- **NEVER** operate a machine when any one of the guards or access panels are open, loose, or removed.
- Make sure the machine is prepared to run in a safe manner by following the machine start information in the Instruction manual.
- Prior to starting the machine, be aware of all personnel in the machine area and make sure everyone is clear.

GENERAL INFORMATION

SAFETY

Safety
Precautions for
Everyone

The following information should be used as a supplement to your company's comprehensive safety policies and procedures.

Your company's policies and procedures supersede the following information if, at any time, the two contradict one another.

- Familiarize yourself with your company's safety and lockout/tagout policies and follow them. Prior to adjusting or servicing the machine, make sure you know which lockout/tagout procedure and safety guidelines apply. When in doubt, get answers from your supervisor before you begin.
- Never reach into, clean, lubricate, adjust, climb on, or work on a moving machine for any reason, even when the machine is only being manually turned over or running at jog speed. ALWAYS follow your company's lockout/tagout policy prior to doing any of the above.
- Electrical shock or unexpected machine movement can cause serious injury or death. For your safety, follow your company's lockout/tagout policy as required for each task.
- Moving components can crush and dismember. Keep hair, body parts, loose clothing, etc. away from moving components.
- NEVER deface or remove factory installed safety signs. If a safety sign is illegible, missing, or damaged in any way, report it to your supervisor. She/he will go through the appropriate procedure to obtain a replacement safety sign.
- Machine mounted blades and knives can cause serious injury, including amputation: keep clear of blades and knives. When supplied, always install/use supplied blade edge and knife guards/shields as stated in the Instruction manual.
- Hot surfaces/liquids may cause burns. Fluids under pressure can be extremely hot.
- Allow ample time for surfaces and system components/fluids to cool before servicing the system. Use an appropriate temperature measuring device (not your hand) to monitor the temperature.

If left around the machine area, tools, spare or loose machine parts, and debris can cause serious injury. Use Loctite products (or equivalent) to secure components that are not used as frequent adjusting devices. Keep the machine area clean. When tools and spare machine parts are not in use, store them in the appropriate locations.

GENERAL INFORMATION

SAFETY

- Lifting heavy components and/or dropping them can cause back injuries, muscle strains, and broken foot bones. Obtain assistance when necessary and use adequate lifting and moving devices properly.
- Tripping and slipping can cause serious injury. Keep walkways clear. Walking surfaces must be free of liquids (oil, grease, ink, coffee, etc.) and other debris.
- Equipment surfaces that are touched by hand (operator interface touch screens, control buttons, levers, etc.) must be free of liquids.
- Disconnect/dump all power sources and dissipate stored energy prior to removing guards and/or access panels. Install and secure all guards and access panels after completing an adjustment, clearing a wrap/jam, and/or servicing the machine. Check affected safety devices to make sure they function properly.
- The only personnel in the general work area should be those operating/servicing the machine. Restrict all others from the area.
- Wear protective clothing (hardhat, eye/face shield, gloves, etc.) as necessary.
- Always wear steel mesh reinforced gloves and arm guards (or equivalent) supplied by your company and use supplied edge guards/shields when handling or working near sharp objects (blades, slitters, knives, etc.).
- Use the proper tools and equipment for the task being performed. Check the tools routinely to make sure they are in proper working condition.
- Follow the procedures in the Instruction manual when starting, adjusting, and servicing the machine and when clearing wraps/jams.
- Check safety devices routinely and replace them at the first indication of defectiveness or failure.
- Prior to starting the machine at the beginning of your shift, walk around it and make sure all guards and other safety devices are in place. Always check for "Do Not Start" tags before starting. **NEVER** start the machine until you are sure that everyone in the area is clear of the machine and aware that it is going to be started.

GENERAL INFORMATION

SAFETY

Electrical Safety

This machine uses electricity of sufficient voltage to cause serious injury or death if mishandled.

NOTE: Drive settings have been adjusted and set at JBT/Prime Equipment Group. Consult the parts documentation containing the machine assembly drawings for drive settings. Some machines may have the settings in the instruction manual. Contact JBT/Prime Equipment Group prior to changing any settings.

- Permit only trained/qualified electrical technicians to work on electrical components, regardless of whether the components are live or dead.
- Know the location of all electrical disconnects, shutoffs, and similar devices and know what each disable. Refer to the machine assembly drawings for this information.
- Keep electrical areas (and the machine in general) dry. Never work on electrical components while standing on or near wet surfaces.
- Always assume a circuit/component is live until proven dead by proper testing. Always test circuits/components with appropriate test equipment (not your fingers).
- Always be on the lookout for electrical wires and components that are frayed, cut, loose, broken, or exposed. Repair/replace such wires and components before machine power is restored and the machine is operated.
- Prior to machine operation, make sure wires, motor plugs, cables and similar devices are securely connected.
- **Never** remove a lockout device or "Do Not Start" tag unless you were the one who put it there. Notify everyone in the machine area when you remove your lockout/tagout that power is going to be restored to the machine.

GENERAL INFORMATION

SAFETY

Flammable/ Hazardous Materials Safety

Flammable/hazardous materials may be present during the converting process. Improperly handled flammable/hazardous materials can cause serious injuries.

NOTE: Material Safety Data Sheets (MSDS) for all flammable/hazardous materials on your machine, and within systems that your machine is equipped with shall be supplied.

- Familiarize yourself with the potential dangers presented by flammable/hazardous materials on/in your machine. The product label, vendor literature, MSDS sheets, and/or the Instruction manual Caution, Warning, and Danger statements state what precautions should be taken to help prevent injury.
- When handling hazardous materials, wear recommended protective clothing (eye/face shield, gloves, apron, etc.) and carefully follow the directions on the label, in the vendor literature, on the MSDS sheets, and/or in the Instruction manual statements.
- Store cleaning fluids and other spare flammable/hazardous materials in a safe place, away from the machine area. This includes rags and other items that have been contaminated with flammable/hazardous material.
- NEVER store flammable/hazardous materials in containers that are not labeled correctly or in containers that are not approved/appropriate for the material being stored within them.
- Restrict smoking and open flame in areas containing flammable/combustible material.
- Keep fire extinguishers fully charged and in convenient locations throughout the machine area. Have the extinguishers checked routinely and make sure they are rated for the type of flammable/hazardous material being used. NEVER return a used or empty extinguisher to its original location until it has been recharged. Have spare, fully charged extinguishers on hand to replace used extinguishers until they have been recharged.
- Dispose of flammable/hazardous materials in accordance with applicable local, state, and federal regulations.

GENERAL INFORMATION

SAFETY

Hazard Warnings

JBT/Prime Equipment Group uses safety signs mounted near or within potentially hazardous machine areas to warn machine users of residual risks in that area (residual risks are those that could not be completely safeguarded for all instances of use). JBT/Prime Equipment Group also uses Caution, Warning, and Danger statements throughout the Instruction manual to alert machine users of potential hazards and instruct them on measures to take to avoid personal injury.

The safety signs on the machine and the Danger, Warning, and Caution statements throughout the Instruction manual use signal words to identify the seriousness of the potential hazard. These signal words are Danger, Warning, and Caution and are defined on the next page.

GENERAL INFORMATION

SAFETY

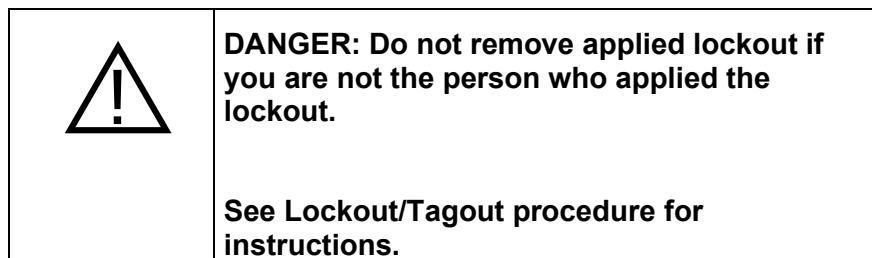
Safety Signs

Machine users must be trained to understand the meaning of these signal words (Danger | Warning | Caution) prior to operating and/or servicing the machine.

If a safety sign is illegible, missing, or damaged in any way, report it to your supervisor. She/he will go through the appropriate procedure to obtain a replacement safety sign.

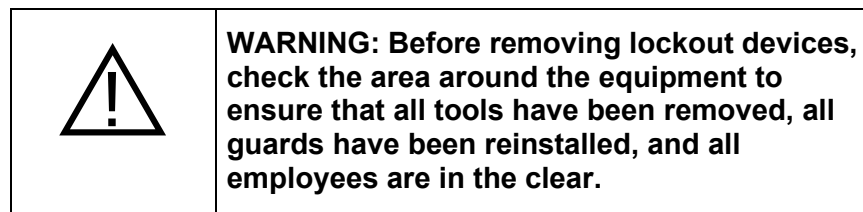
The **DANGER** statement indicates an immediate hazard or unsafe practice that, if ignored, **will** result in **serious injury or death**.

Example:



The **WARNING** statement indicates a potential hazard or unsafe condition that, if ignored, **could** result in **serious injury**.

Example:



GENERAL INFORMATION

SAFETY

The **CAUTION** statement indicates a potential hazard or unsafe condition that, if ignored, **might** result in **minor injury and/or machine damage**.

Example:

	CAUTION: For continued protection against possible damage to the machine, check the area for debris before starting up the machine.
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GENERAL INFORMATION

SAFETY

Safety During Machine Installation & Disassembly

Machine installation/disassembly presents hazards and unique challenges to those unfamiliar with installation/disassembly techniques. If specified, experienced service technicians will install the machine. If you select an alternative installation company or decide to install the machine yourself, JBT/Prime Equipment Group recommends having our service technician(s) check installation and initially start/adjust the machine.

The following information should be used as a reference guide when installing/disassembling machines. It should not be considered a substitute for experience or assistance.

- Know and comply with applicable federal, state, and local codes regarding machine installation/disassembly.
- Rope/block off the area where the machine is going to be installed/disassembled. Allow only authorized personnel directly related with installation/disassembly within the area.
- Make sure everyone working in the installation/disassembly area is supplied with and wears all necessary protective clothing (hardhats, steel toed footwear, gloves, etc.).
- Review the installation drawing package prior to machine installation. When applicable, the drawings have weights and lift points called out on them.
- If you have questions, get answers before you start to install the machine. Keep the installation drawings at the installation site for quick reference during the installation process.

GENERAL INFORMATION

SAFETY

- Use lifting and moving equipment (overhead cranes, hoists, straps, chains, forklifts, etc.) that meet or exceed each machine section weight. Make sure the lifting and moving equipment is in proper working condition.
- Do not rush. The equipment is only installed once. Doing it safely, thoroughly, and correctly will prevent accidental injury and help ensure satisfactory machine operation.
- Keep the work area clean. Remove debris from the area often. Put away tools that are not being used.
- Install all guards and access panels that may have been removed for shipping purposes.
- **NEVER** defeat or remove any of the safety devices. Each one performs a specific function and all of them are provided for your protection.
- Prior to applying power to the machine, walk around the machine making sure all guards, access panels, interlocks, lifelines, and other safety devices are installed/functional and that all personnel in the area are clear of the machine.

GENERAL INFORMATION

SAFETY

Machine Guarding

JBT/Prime Equipment Group machines are equipped with operator access guards, maintenance access guards, and fixed guards to protect machine users from potential hazards (nips, meshed gears, rotating shafts, spinning blades, etc.).

NOTE: Only authorized personnel should be allowed to open or remove any guard. All guards must be properly closed or reinstalled prior to resuming operation.

Operator Access Guards

Operator access doors and guards are either electronically interlocked (restricts entry to the area until machine components reach zero speed, with or without timer at zero speed) or interlocked (initiates machine shutdown if the machine is in the run mode and a door or guard is opened).

Prior to opening the operator access doors or guards, it is the responsibility of the person that will open the door or guard to place the appropriate lockout/tagout on the machine following your company's lockout/tagout policy.

Both types of interlocks restrict the machine from starting in the run mode if an operator access door or guard is open.

In certain instances, machine jogging is permitted with an operator access door or guard open. This can only be done using the jog control in the immediate vicinity of the opened operator access door or guard. This ensures that machine movement can only be initiated by a person adjacent the open door/guard and that all potential hazards in the immediate area can be seen by her/him.

NOTE: The normal stop control should always be used to trigger all non-emergency machine stops. When the machine is in the run mode, interlocked doors or guards should only be opened to emergency stop the machine when saving life or limb.

GENERAL INFORMATION

SAFETY

Maintenance Access Guards

Maintenance access guards (example: doors to drive components) are typically hinged or removable. They require a tool to open them. Since use of a tool is required, they are not typically interlocked.

Prior to using the tool to open or remove the access panel or guard, it is the responsibility of the person that will open/remove the panel or guard to place the appropriate lockout/tagout on the machine following your company's lockout/tagout policy. Close/replace the access panel or guard prior to removing the lockout/tagout.

Fixed Guards

Fixed guards are used to cover machine areas that are infrequently accessed. These types of guards are not interlocked, hinged; or in any way, designed to allow operator or service technicians easy access to the area.

Fixed guards should only be removed by company authorized personnel (maintenance personnel, service technicians, etc.).

Prior to removing a fixed guard, it is the responsibility of the person that will remove the fixed guard to place the appropriate lockout/tagout on the machine following your company's lockout/tagout policy. Install all removed fixed guards prior to removing the lockout/tagout.

GENERAL INFORMATION

SAFETY

Pressurized System Safety

The machine could be equipped with pneumatic and hydraulic systems to move and control machine components. These systems are typically under pressures high enough to cause serious injury if mishandled.

- Obtain company training/authorization prior to servicing a pressurized system.
- Use the proper tools and equipment for the task being performed. Check the tools routinely to make sure they are in proper condition.
- Fluids under pressure can be extremely hot. Allow ample time for the system fluid/components to cool before servicing the system. Use an appropriate temperature measuring device (not your hand) to monitor the temperature.
- Lockout/tagout the machine and relieve stored energy in the pressurized system following your company's lockout/tagout policy prior to servicing/disassembling any portion of the system.
- Never operate or pressurize a system that has worn/damaged components. Never adjust pressurized systems beyond the recommended setting/range. The Machine Setting section of the Instruction manual provides base pressure settings for all pressurized systems that are not run at line pressure. These base settings could require minimal adjustment to best suit actual running conditions. Do not increase/decrease machine stopping times without consulting JBT/Prime Equipment Group.

GENERAL INFORMATION

WARNING LABELS AND LOCATIONS



SAFETY STICKER MAY COME OFF DUE TO USE OF HIGH-PRESSURE WATER. IF A STICKER IS MISSING IT MUST BE REPLACED IMMEDIATELY.

PART NO.	DESCRIPTION	QTY.	IMAGE
19883	SAFETY STICKER, WARN, VOLTAGE, HORIZONTAL	1	

GENERAL INFORMATION

SYSTEM DESCRIPTION

System Description

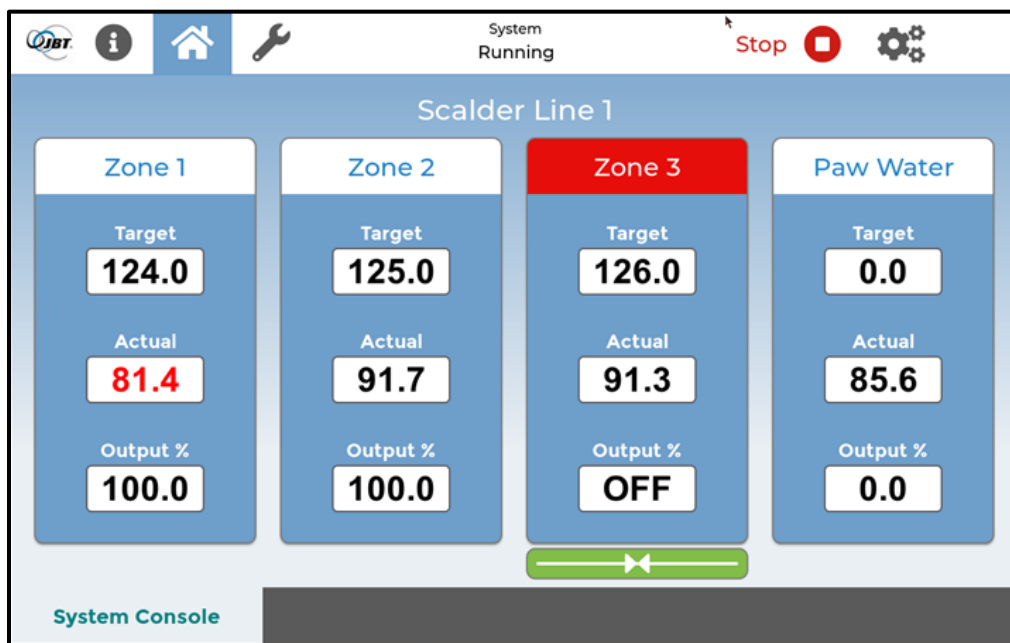
The **JBT Scalders System** monitors and maintains a consistent temperature inside a scalders tank by continuously adjusting a steam valve. The system monitors the temperature inside a tank with an RTD temperature probe. If the temperature inside the tank is too low, the system will increase the amount of steam by opening the valve further. If the temperature is too high, the system will decrease the amount of steam by further closing the valve. The system supports monitoring and maintaining the temperature of up to 6 scalders tanks with a single JBT Controller.

Hardware Options

1 – 6 Scalders Tanks

Optional Redundant Temperature Sensors

Digital or Analog Water Level Sensors to control water valves and/or alarms.



JBT Efficiency Management System

Most counter installations also include JBT EMS (Efficiency Management System) software that runs on a Windows computer in a nearby office. The JBT EMS software continuously uploads data during the day and stores it in a Microsoft SQL Server database. The program makes this data available to users via reports, charts, status screens, printing, automatic report emailing, alarms and many other useful features. There is also a client version of the software that allows users to run the program on their own computer in their office.

GENERAL INFORMATION

EQUIPMENT TRAINING RECORD

All Sanitation and Maintenance personnel must be trained in the following areas. Prime Equipment Group recommends you keep a record of personnel training. Prime Equipment Group is willing to assist with training, following a request to do so. Please feel free to call Prime Equipment Group for additional information.

- Installation and product application
- Disassembly and sanitation
- Instruction and parts manual
- Daily periodic maintenance
- Wearing parts
- Assembly and setup for operation
- Safety features and precautions
- Troubleshooting
- Recommended spare parts inventory
- Routine preventative maintenance schedule

GENERAL INFORMATION

LIMITED WARRANTY

Products purchased from Manufacturer are warranted only by the Manufacturer's warranty that the Product provided herein is of their standard quality and operable and that the Product will be free from defects in workmanship and material. The warranty period, for Product sold by Manufacturer and not expressly made subject to a different warranty, is one hundred twenty (120) days from date of receipt by Customer. There is no insurance against or warranty of any type for production loss due to late or damaged Product shipments. The Manufacturer makes no express or implied statutory warranties other than as expressly set forth herein.

Manufacturer shall for a period of one hundred twenty (120) days from the date any Product sold hereunder is received by Customer, repair or replace at its option, free of charge and F.O.B. point of manufacture, any nonconforming or defective parts or Product that upon inspection by Manufacturer, are deemed to be nonconforming or defective. Manufacturer shall, however, only be required to repair or replace such parts or Products if Customer (i) owned the nonconforming or defective parts or Product continuously from the original date of delivery, (ii) stored and maintained such parts and Product in accordance with Manufacturer's recommendations and standard industry practices and (iii) notified Manufacturer in writing of any nonconformance or defective parts or Product within fifteen (15) days following the date the nonconformity or defect became reasonably apparent. Further the obligations of Manufacturer hereunder do not apply to the repair or replacement of Products within the aforementioned warranty period for (a) misuse, negligence or accident, (b) normal maintenance services, (c) normal replacement of service items in connection with such service, (d) normal deterioration due to wear and tear, (e) use of non-manufacturer (OEM) replacement parts. The remedy provided in this Warranty shall be the sole and exclusive remedy of Customer in the event of defective or nonconforming parts or Product, and shall be void if the Product has been altered in any manner or has been moved from its original location without the written approval of the Manufacturer. The repair or replacement of any parts or Products

under the foregoing provision does not extend the warranty beyond the warranty period described above. All electric motors, controls and other electrical services and their component parts, hydraulic motors, controls and other hydraulic services and their component parts, pneumatic controls, pneumatic services and other pneumatic components, bearings, gearboxes etc. are not included within this Warranty; the warranty with respect to these parts being limited to the manufacturer of these parts, a copy of which may be obtained from Manufacturer upon request. Customer will be responsible for the shipping/packaging costs associated with the safe and expedient return of warranted parts. The Warranty can only be claimed if genuine Manufacturer spare parts are used for repairs and maintenance of the Product.

MANUFACTURER'S UNDERTAKINGS AS CONTAINED IN THIS WARRANTY AS SPECIFIED ABOVE SHALL BE IN LIEU OF ANY OTHER GUARANTY OR WARRANTY, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTY OF MERCHANTABILITY OR IMPLIED WARRANTY OF FITNESS FOR A PARTICULAR PURPOSE. IN ADDITION, IT IS EXPRESSLY AGREED BETWEEN THE PARTIES THAT THE MANUFACTURER HAS NOT MADE ANY REPRESENTATIONS OR WARRANTY AND THERE IS NO AGREEMENT THAT THE PRODUCT PURCHASED HEREUNDER WILL MEET ANY PARTICULAR

STANDARD UNLESS CUSTOMER HAS SUPPLIED MANUFACTURER WITH WRITTEN SPECIFICATIONS OF THE PRODUCT AND THE MANUFACTURER HAS AGREED IN WRITING THAT THE PRODUCT WILL MEET SPECIFIC STANDARDS SPECIFIED BY THE CUSTOMER. CUSTOMER UNDERSTANDS AND ACKNOWLEDGES THAT THERE ARE NO WARRANTIES WHICH EXTEND BEYOND THE TERMS HEREOF UNLESS OTHERWISE NOTED IN PURCHASE AGREEMENT.

NO CLAIM OF ANY KIND, INCLUDING, BUT NOT LIMITED TO CLAIMS OF NEGLIGENCE AND BREACH OF CONTRACT, WHETHER AS TO PRODUCT DELIVERED OR SERVICES RENDERED OR FOR NON-DELIVERY OF PRODUCT OR NON-PERFORMANCE OF SERVICES SHALL BE GREATER THAN THE PURCHASE PRICE OF THE PRODUCT OR SERVICES IN RESPECT OF WHICH THE CLAIM IS MADE. WITHOUT LIMITING THE FOREGOING, MANUFACTURER'S LIABILITY HEREUNDER SHALL, UNDER NO CIRCUMSTANCES, EXCEED THE PURCHASE PRICE SET FORTH IN THIS QUOTATION.

MANUFACTURER SHALL, IN NO EVENT, BE LIABLE FOR ANY INCIDENTAL, CONSEQUENTIAL, INDIRECT OR SPECIAL DAMAGES, LOSS OF BUSINESS PROFIT, PERSONAL INJURY, BUSINESS INTERRUPTION AND LOSS OF BUSINESS, EVEN IF MANUFACTURER HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES AND ANY CLAIM BY CUSTOMER TO SUCH DAMAGES, WHETHER BASED ON TORT, CONTRACT OR OTHERWISE IS HEREBY EXPRESSLY WAIVED AND EXCLUDED.

OPERATIONS AND INSTRUCTIONS

SPECIFICATIONS

SN4448 Utilities.

Electrical.

- Customer is responsible for providing an electrical drop to the controller and terminating it.
 - Dedicated 120VAC Single Phase 60 HZ 6 Amps.
- Customer is responsible for running the JBT Prime supplied 6 conductor 20AWG communication wire from each RTD back to the controller and from each Transducer Assembly back to the controller. The JBT Marel tech will be responsible for terminating the wire.
 - 24VDC, 4-20mA signal.

Plumbing.

- Customer is responsible for plumbing/attaching the RTDs into the various tanks and is responsible for providing any needed thermowells to thread the RTDs into their tanks.
 - 1/2" NPT connection.

Air.

- Customer is responsible for providing an air drop with clean dry air to the quantity 2, transducer boxes and for terminating into each box.
 - 1/4" push lock connection 100PSI
- Customer is responsible for providing and running 1/4" poly flow air tubing from the transducer boxes to each of the steam valves.
 - 1/4" push lock connection on the transducer assemblies

Network/Ethernet.

- Customer is responsible for running shielded CAT 6 ethernet cable from the controller to their nearest network switch.
- PLEASE NOTE CUSTOMER HAS PROVIDED THE FOLLOWING.
- **Static IP Address Controller**
 - IP: 10.116.70.153
 - SN: [255.255.255.0](#)
 - GW: 10.116.70.254
 - DNS1: [167.110.87.115](#)
 - DNS2: [167.110.212.4](#)

OPERATIONS AND INSTRUCTIONS

SPECIFICATIONS

- Customer is responsible for providing remote VPN access to the controller to JBT Prime Principal Software Engineer Chris Freeman chirs.freeman@jbt.com.

Mounting

- Customer is responsible for all labor and material needed to mount the equipment and for providing and running any needed conduit.

JBT System Navigation Site Map






03/14/23 09:39
Stopped

Start 



- **Info/About**
 - Company
 - Contact Info
 - Software Version






03/14/23 09:39
Stopped

Start 



- **Home**
 - **System Console**
 - Scalders
 - Communication Errors






03/14/23 09:39
Stopped

Start 



- **Tools**
 - **Operator**
 - Target Temperatures
 - **Maintenance**
 - Sensor Setup
 - Raw Temperature Data
 - **Hardware**
 - Hardware Test Screens
 - I/O List






03/14/23 09:39
Stopped

Start 

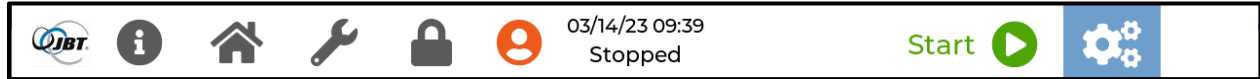


- **Advanced Settings**
 - **Advanced**
 - Min/Max Temperatures
 - Control Temperatures
 - Controller
 - Security
 - **Admin**
 - General
 - JBT Adv Admin (JBT Technician)

OPERATIONS AND INSTRUCTIONS

NAVIGATION BAR

Navigation Bar



The Navigation Bar is the 1st layer of navigation. The selected icon is shown in blue.

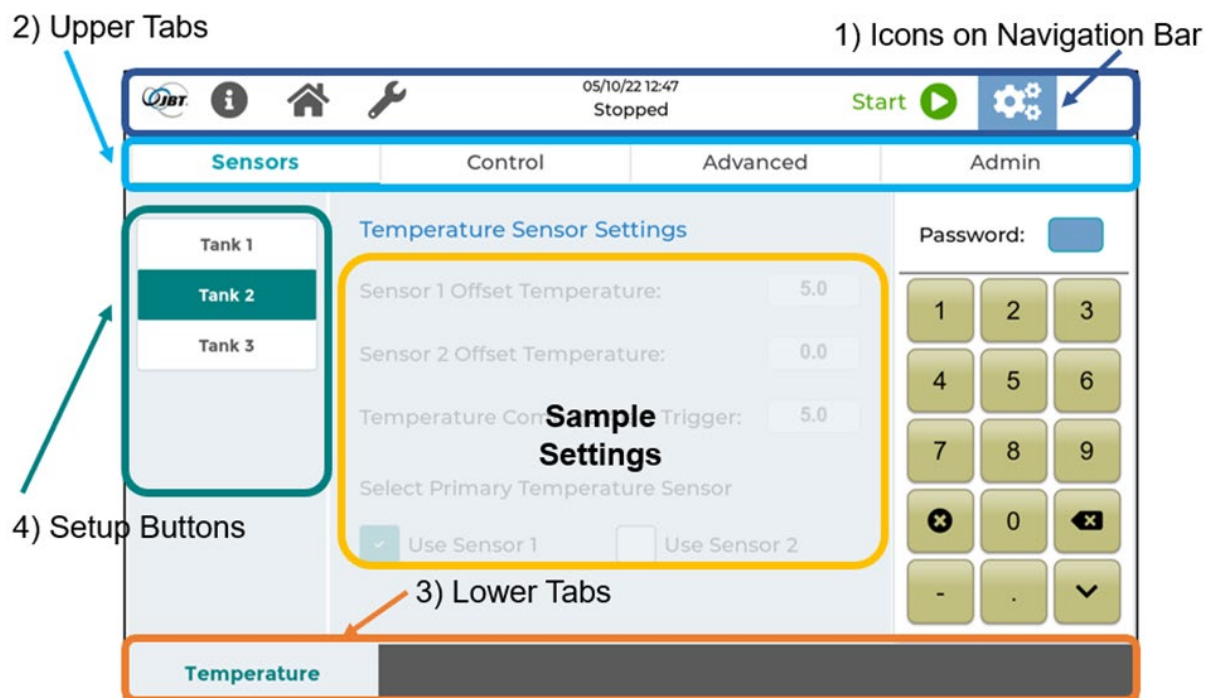
Icon	Name	Description
	Info (About)	Info about the company, contact info, program name and version.
	Home	The home screen shows the system in action. Startup and time-outs default to this screen.
	Operator System	The Tools section contains tools used by operators and their supervisors.
	Logged on Security Level Touch to manually logout.	- Operator (Green): not shown if operator passwords are turned off. - Maintenance (Purple) - Advanced (Orange) - Super User (Red) Used during installation.
05/04/22 15:45 Running	System State	The center of the navigation bar shows the current time and system state
Start	Start	Start the System. There is an option on Controller Setup to Autostart after Power-Up.
Stop	Stop	Stop the System. Passcord required to stop the system.
	Advanced Settings	The Settings section contains advanced System Settings, Security Settings, and Admin Settings.

JBT System Navigation

Navigation Overview

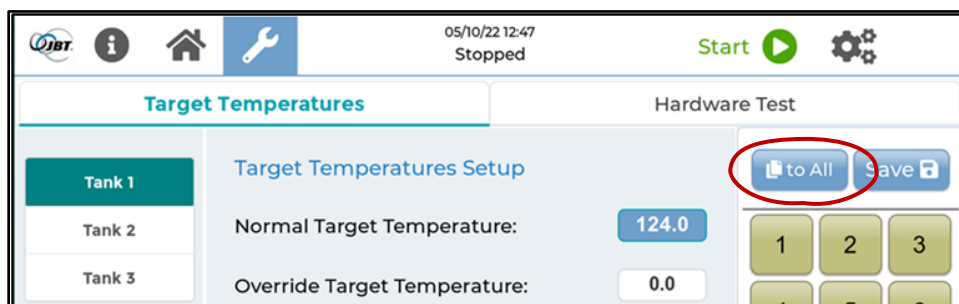
Navigation occurs in the following steps. Some screens may not use each of these steps (for example, the screen may not need setup buttons on the left), however the steps are always in this order.

Sample navigation steps to a setup screen:



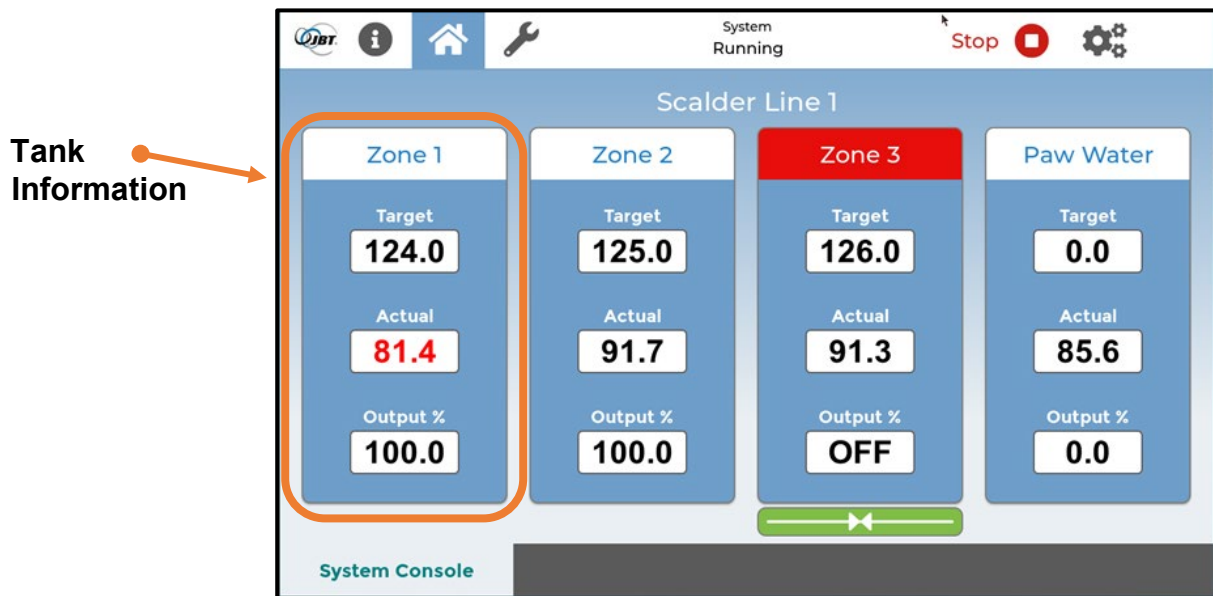
Settings (Cogs) – **Sensors** (Upper Tab) – **Temperature** (Lower Tab) – **Tank 2** (Setup Button)

Copy to All button copies the values of the 1st entity to other entities.



System Console

The JBT Scalders System displays the System Console Home screen and the current temperature data.



The **Target** value is the desired temperature for the system and the **Actual** value is the current temperature.

The **Output %** only applies to systems that are controlling temperature sensors and displays the current percentage that the steam or mixing valve is open. 'OFF' means the steam was disabled manually by a pushbutton or disabled automatically. N/A for Not Applicable is used by systems that do not control water temperatures.

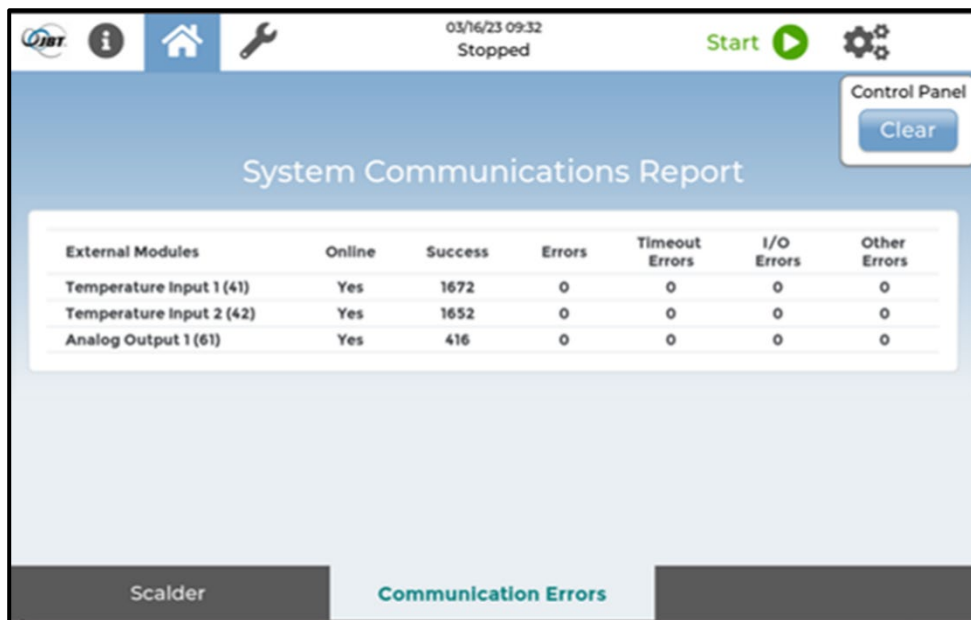
Water Level Sensors and Alarms

Adding sensors to control the water level is an available feature. Digital or Analog sensors detect the level of the water and can open a water valve and/or set alarms based on the water level. In the example above, Zone 3 has reached a critical low water level, and as a result, the following actions are shown on the screen:

1. The Red banner indicates that water level has dropped below the sensor designated as the 'critical low' sensor. Other colors, such as yellow and orange, are used for upper sensors as configured for your system.
2. The output is 'OFF' meaning the steam valve is closed due to low water.
3. The green open valve symbol means the water valve is open to add water.
4. Some systems are configured with alarms, such as strategically placed lights and/or horns.

System Console – Communication Errors

The Communications Report displays the status of each Advantech module used with the Scalder System. Your installation may show a different number of modules. The first two modules shown above are Temperature RTD Inputs and the third module is the Analog Output module controls the temperature of the systems.



External Modules	Online	Success	Errors	Timeout Errors	I/O Errors	Other Errors
Temperature Input 1 (41)	Yes	1672	0	0	0	0
Temperature Input 2 (42)	Yes	1652	0	0	0	0
Analog Output 1 (61)	Yes	416	0	0	0	0

The System Communications Report displays the status of each Advantech module used with the Scalder System. Your installation may show a different number of modules. The first two modules shown above are Temperature RTD Inputs and the third module is the Analog Output module controls the temperature of the systems.

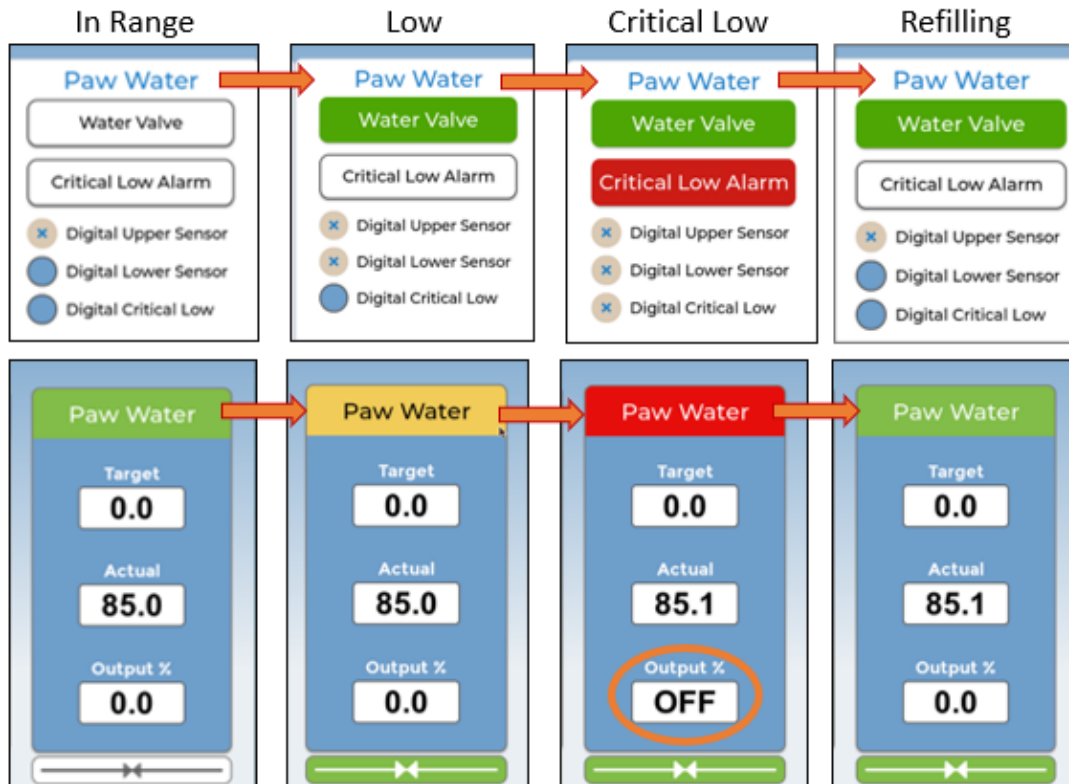
The Online column shows whether we are currently communicating with the module as Yes or No. The Success and Errors columns show how many communications with the module were successful and how many had errors. The last three columns show a breakdown of the types of errors.

Digital Water Level Sensors (Optional)

The JBT Scalders System supports digital water level sensors placed at the following locations in the tank:

- **Upper Sensor:** The upper range of the desired water level. This sensor is used to turn off a water valve when the water level reaches this sensor.
- **Lower Sensor:** The lower range of the desired water level. This sensor is used to set a light/alarm and/or turn on a water valve. The system console shows green for any water level above this sensor, and yellow when the water level falls below this sensor.
- **Critical Low Sensor:** Detects when the water level is critically low. This sensor is used to set an alarm/light. The alarm output can also be used to send a low-voltage signal to auxiliary devices such as horns or water pumps. The system console shows red and the tank's steam outputs are disabled while the critical low state is in effect.
- **Ending the Critical Low Sensor Alarm State:** Once the critical low sensor alarm has been activated, the critical low alarm state continues until the sensor has been submerged continuously for 30 seconds. Any movement of the water that uncovers the sensor restarts the timer. This ensures that the sensor is securely submerged before turning off the critical alarm and re-enabling the steam output.

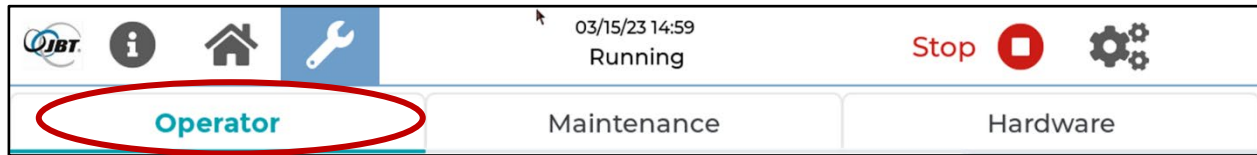
The sample sequence below shows system activity moving from normal water range to critical low and then refilling toward the upper sensor.



OPERATIONS AND INSTRUCTIONS

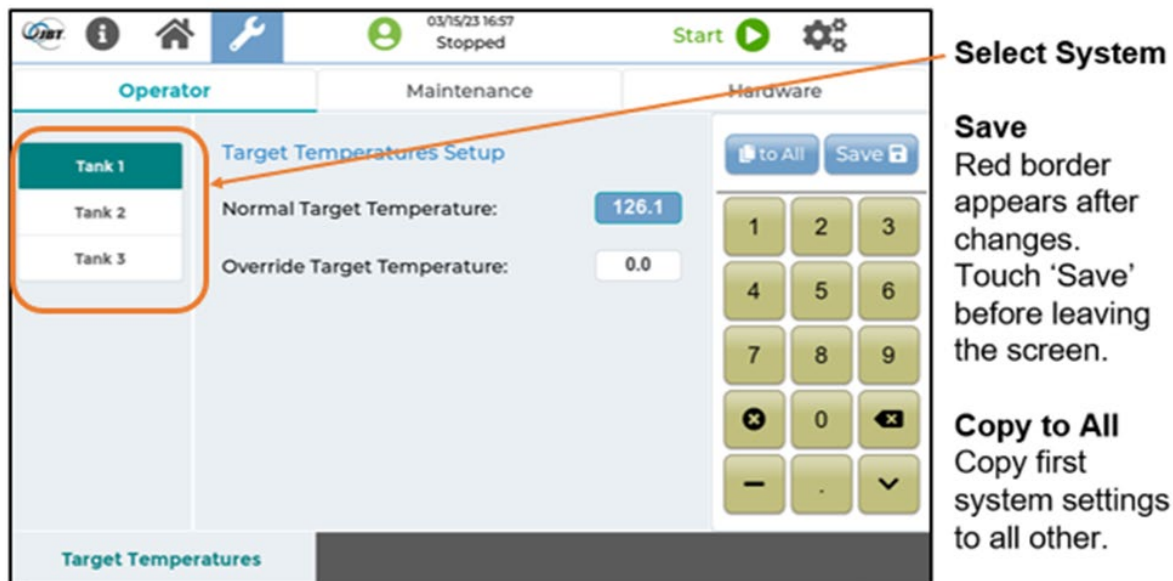
TOOLS (WRENCH) - TARGET TEMPERATURES

Tools (Wrench) – Operator - Target Temperatures



Navigation: Tools (Wrench) – Operator (Upper Tab) - Target Temperatures (Lower Tab)

Security Level: Level 1 Operator Password or above.



Select System

Save

Red border appears after changes. Touch 'Save' before leaving the screen.

Copy to All

Copy first system settings to all other.

The **Normal Target Temperature** is the temperature that the system should optimally be at all the time.

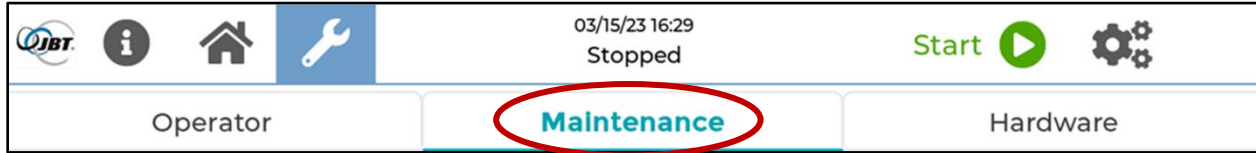
The **Override Target Temperature** is a second target temperature that will be used when the Override Target Temperatures pushbutton is pressed.

Temperatures are specified in degrees Fahrenheit.

OPERATIONS AND INSTRUCTIONS

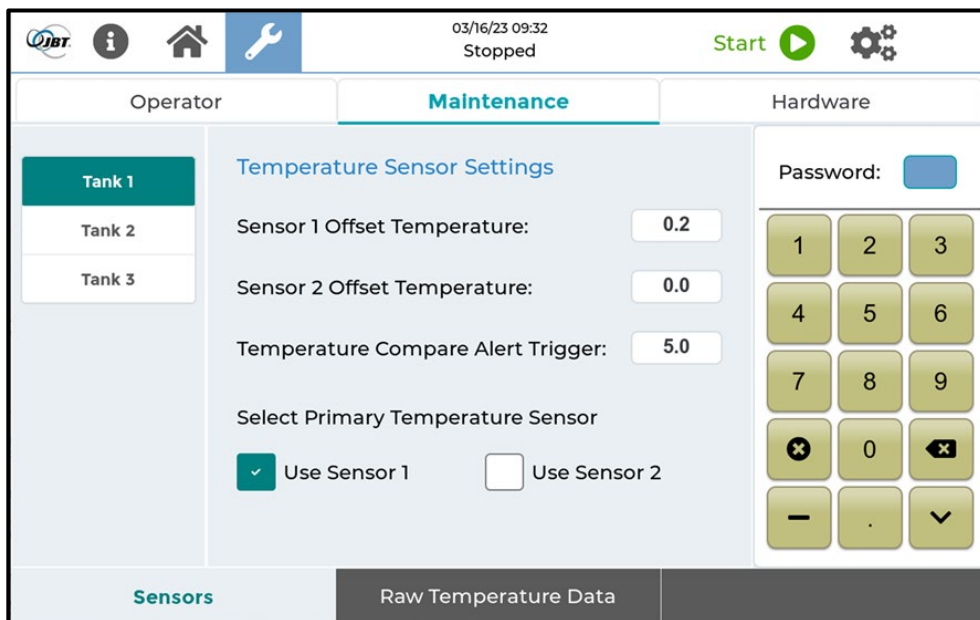
TOOLS (WRENCH) - SENSORS

Tools (Wrench) – Maintenance – Sensors



Navigation: Tools (Wrench) – Maintenance (Upper Tab) – Sensors (Lower Tab)

Security Level: Level 2 System Password or above.



The **Offset Temperature** is the offset in degrees that will be added to or subtracted from the measured temperature from the sensor. Factors such as the length of cable between the JBT Controller and the RTD temperature probe can cause the measured temperature to be skewed either high or low. The offset temperature will account for this by adjusting the measured temperature to obtain an accurate actual temperature. Configure the offset temperature for sensor 1 and for sensor 2 if using redundant temperature sensors.

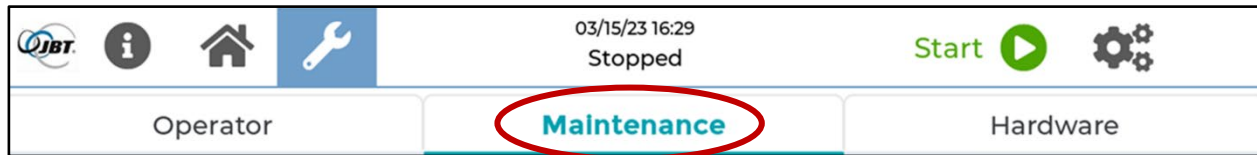
The **Temperature Compare Alert Trigger** compares the temperature readings from the two temperature sensors if using redundant temperature sensors. If the difference is greater than the Alert Trigger value, then the Actual Temperature will be displayed in red on the Home screen and the General Alarms output will be turned on. Configure the Temperature Compare Alert Trigger for each system as desired.

The **Primary Temperature Sensor** is the sensor whose value will be used for controlling the temperature of the system. The primary sensor's temperature is also the value that will be displayed on the Home screen. Touch the box to the left of which sensor you want to be primary for each system.

OPERATIONS AND INSTRUCTIONS

TOOLS (WRENCH) - RAW TEMPERATURE DATA

Tools (Wrench) – Maintenance – Raw Temperature Data



Navigation: Tools (Wrench) – Maintenance (Upper Tab) – Raw Temp Data (Lower Tab)

Security Level: No Passcode required.

The 10 most recent raw temperature values from both RTDs are displayed below each system. Your plant may only have 1 RTD for a system which means that only 1 value would be displayed.

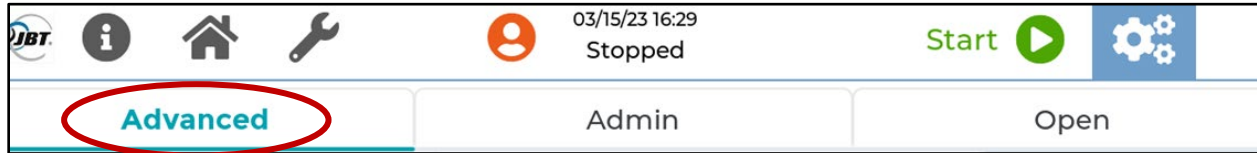
Tank 1	Tank 2	Tank 3	Paw	Rail	Makeup
125.06 125.10	127.30 127.13	127.97 128.30	145.20 145.13	120.17 120.80	121.00 119.43
125.07 125.10	127.31 127.14	127.97 128.30	145.21 145.14	120.17 120.80	121.01 119.44
125.07 125.10	127.31 127.15	127.96 128.30	145.21 145.15	120.16 120.80	121.01 119.45
125.06 125.10	127.31 127.15	127.96 128.32	145.21 145.15	120.16 120.82	121.01 119.45
125.06 125.12	127.32 127.15	127.96 128.32	145.22 145.15	120.16 120.82	121.02 119.45
125.06 125.12	127.33 127.18	127.96 128.30	145.23 145.18	120.16 120.80	121.03 119.48
125.06 125.10	127.33 127.09	128.01 128.41	145.23 145.09	120.21 120.91	121.03 119.39
125.11 125.21	127.29 127.17	128.01 128.35	145.19 145.17	120.21 120.85	120.99 119.47
125.11 125.15	127.33 127.17	127.97 128.35	145.23 145.17	120.17 120.85	121.03 119.47
125.07 125.15	127.31 127.13	127.99 128.37	145.21 145.13	120.19 120.87	121.01 119.43

The raw data will be continually updated with the most recent values at the bottom.

OPERATIONS AND INSTRUCTIONS

ADVANCED SETTINGS - MIN/MAX TEMPS

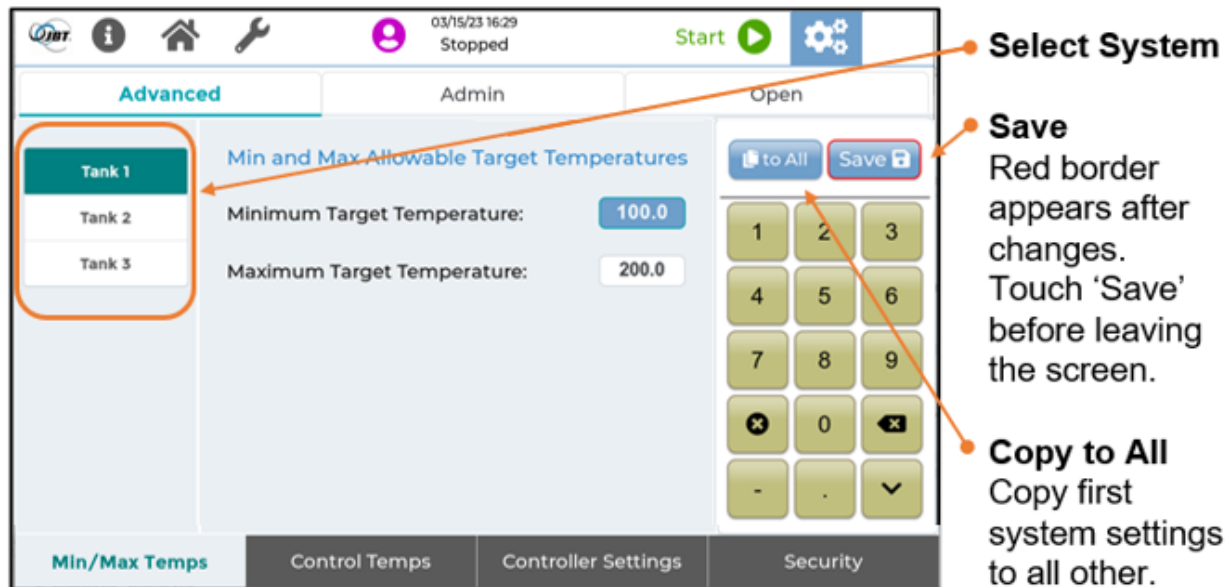
Advanced Settings (Cogs) - Advanced – Min/Max Temps



Navigation: Adv Settings (Cogs) – Advanced (Upper Tab) – Min/Max Temps (Lower)

Security Level: Level 3 Advanced Password or above

The names of the systems that have been installed at your plant will be displayed on the left side of the screen.

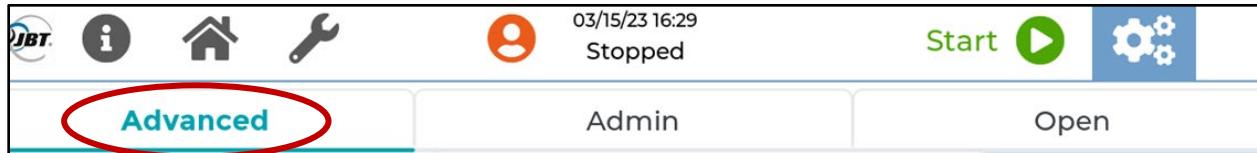


The **Minimum Target Temperature** is the lowest temperature that can be configured on the Target Temperatures Setup screen that is access from the Tools (Wrench) icon.

The **Maximum Target Temperature** is the highest temperature that can be configured on the Target Temperatures Setup screen.

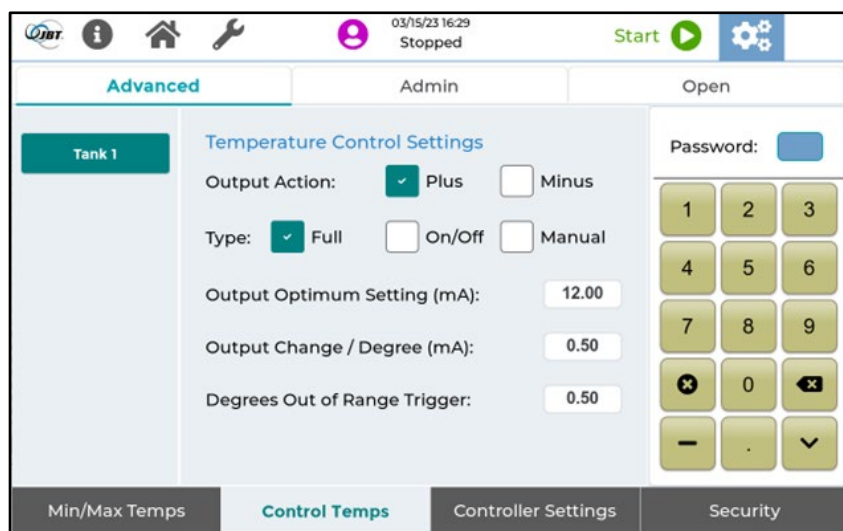
These values prevent someone from setting a target temperature lower or higher than is appropriate for the selected system. Temperatures are specified in degrees Fahrenheit.

Advanced Settings (Cogs) - Advanced – Control Temps



Navigation: Adv Settings (Cogs) – Advanced (Upper Tab) – Control Temps (Lower)

Security Level: Level 3 System Password or above.



The **Output Action** can be set to either **Plus** or **Minus** and indicates whether an increase in the Output value causes a plus (positive) change in temperature or causes a minus (negative) change in temperature. The majority of temperature control systems have a plus output action. Scaldex tanks in a poultry processing plant typically use steam or hot water to change the temperature in a tank. With steam or hot water an increase in the output value further opens a valve causing more steam or hot water to enter the tank. This results in a plus (positive) change to the temperature in the tank. Touch the box to the left of either **Plus** or **Minus** if it is necessary to change the **Output Action** setting. The box to the left of the new desired output action will turn green and the old output action will change back to white.

The **Control Type** can be set to either **Full**, **On/Off** or **Manual** for each system. The Control Type tells the JBT Scaldex System what type of control algorithm to use for maintaining the Actual temperature as close as possible to the Target temperature. The 2 different Control Types will be discussed in detail below.

The Output Action and Control Type settings must be configured for each system separately. When finished configuring settings for all systems, touch the **Save** button.

Full Control

The Full control type that is recommended for Scalders Tanks that use steam (or hot water) to change the temperature in the tank. The control for the steam (or hot water) is usually done by opening and closing a valve as necessary to either increase or decrease the amount of steam entering the tank. In the JBT Scalders System this is done via a 4 to 20 milliamp output signal from the JBT Controller. At 4 milliamps the steam valve will be 100% closed and at 20 milliamps the steam valve will be 100% open. The Full control type requires configuring the 3 settings discussed below.

The screenshot shows the JBT Scalders System Control interface. At the top, there is a status bar with the JBT logo, a home icon, a wrench icon, the date and time '09/26/22 11:38', the status 'Stopped', a 'Start' button with a play icon, and a settings gear icon. Below this is a navigation bar with four tabs: 'Sensors', 'Control' (selected), 'Advanced', and 'Admin'. The 'Control' tab is active, showing 'Temperature Control Settings' for 'Tank 1'. On the left, there is a sidebar with 'Tank 1' and 'Tank 2' buttons. The settings for Tank 1 are as follows: 'Output Action' is set to 'Plus' (checked) with 'Minus' as an option; 'Type' is set to 'Full' (checked) with 'On/Off' and 'Manual' as options; 'Output Optimum Setting (mA)' is set to '12.00'; 'Output Change / Degree (mA)' is set to '0.50'; and 'Degrees Out of Range Trigger' is set to '0.50'. On the right side of the settings panel, there is a 'Password:' field and a numeric keypad with buttons for digits 1-9, 0, a decimal point, and a down arrow. The bottom of the interface shows a 'Temperature' label and a dark grey bar.

- **Output Optimum Setting** (milliamps) is the output value that usually results in the actual temperature being maintained fairly close to the desired target temperature. This value can be determined experimentally during production by using the **Manual** control type and changing the manual output value until the actual temperature is close to the desired target temperature.
- **Output Change / Degree** (milliamps) tells the system how many milliamps to increase or decrease the output signal for every degree that the actual temperature is different from the desired target temperature. The necessary milliamp increase or decrease value will then be added to the **Output Optimum Setting** to obtain the actual value outputted to the steam valve. The resulting value is 4 - 20 milliamps..
- **Degrees Out of Range Trigger** specifies the number of degrees the actual temperature is different from the target temperature that will trigger an out of range condition. When the actual temperature is lower than the desired set temperature by the specified number of degrees then the system will automatically open the steam valve 100% until the temperature increases to within the out of range trigger. When the actual temperature is higher than the set temperature by the specified number of degrees then the system will automatically close the steam valve 100% until the temperature decreases to within the out of range trigger.

OPERATIONS AND INSTRUCTIONS

ADVANCED SETTINGS - CONTROL TEM S

On/Off

The On/Off control type uses the following settings to turn a valve open or closed to maintain the correct temperature.

The screenshot displays the 'Advanced' settings for 'Tank 1' under the 'Control Temps' tab. The 'Temperature Control Settings' section includes the following options:

- Output Action:** ☒ Plus, ☐ Minus
- Type:** ☐ Full, ☒ On/Off, ☐ Manual
- Degrees Out of Range Trigger Low:** 2.00
- Degrees Out of Range Trigger High:** 2.00
- The Values Below Are Specified in Seconds**
- In Range Low:** On: 5.00, Off: 5.00
- In Range High:** On: 5.00, Off: 5.00

A numeric keypad is located on the right side of the settings panel, and a 'Save' button is at the top right of the settings area.

- **Degrees Out of Range Trigger Low** When the actual temperature is lower than the desired set temperature by the specified number of degrees then the system will automatically open the steam/hot water valve until the temperature is above the low trigger value.
- **Degrees Out of Range Trigger High** When the actual temperature is higher than the set temperature by the specified number of degrees then the system will automatically close the steam/hot water valve until the temperature decreases to below the out of range trigger.
- **In Range Low** The number of seconds to cycle the steam/hot water valve on/off while the temperature is between the Low Range Trigger and the target temperature. The purpose is to increase the water temperature without overshooting the target temperature.
- **In Range High** The number of seconds to cycle the steam/hot water valve on/off while the temperature is between the target and the high range trigger temperature. The purpose is to decrease the water temperature without dropping below the target temperature.

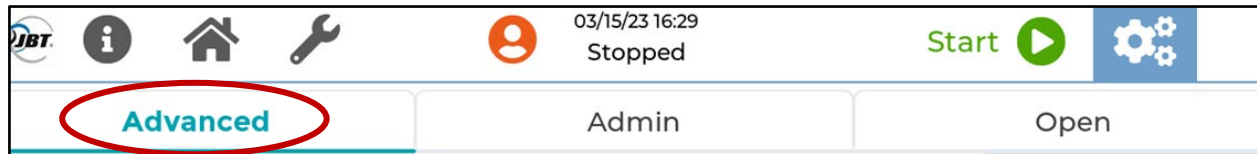
Manual Control

Manual control is generally only used to determine what the Output Optimum Setting is when using the JBT control type. The Manual control type only has 1 setting to configure as shown on the screen at the top of the next page.

The screenshot displays the 'Advanced' settings for 'Control Temps'. Under 'Temperature Control Settings' for 'Tank 1', the 'Output Action' is configured to 'Plus' (indicated by a checked checkbox) and 'Minus' (unchecked). The 'Type' is set to 'Manual' (checked), with 'Full' and 'On/Off' (unchecked) also listed. The 'Output Setting (mA)' is set to '12.00'. A numeric keypad is visible on the right side of the screen, and a 'Save' button is located at the top right. The bottom navigation bar shows 'Min/Max Temps', 'Control Temps' (highlighted), 'Controller Settings', and 'Security'.

- **Output Setting** value for the **Manual** control type is specified in milliamps and is the fixed value that will be used continuously regardless of how close the actual temperature is to the target temperature.

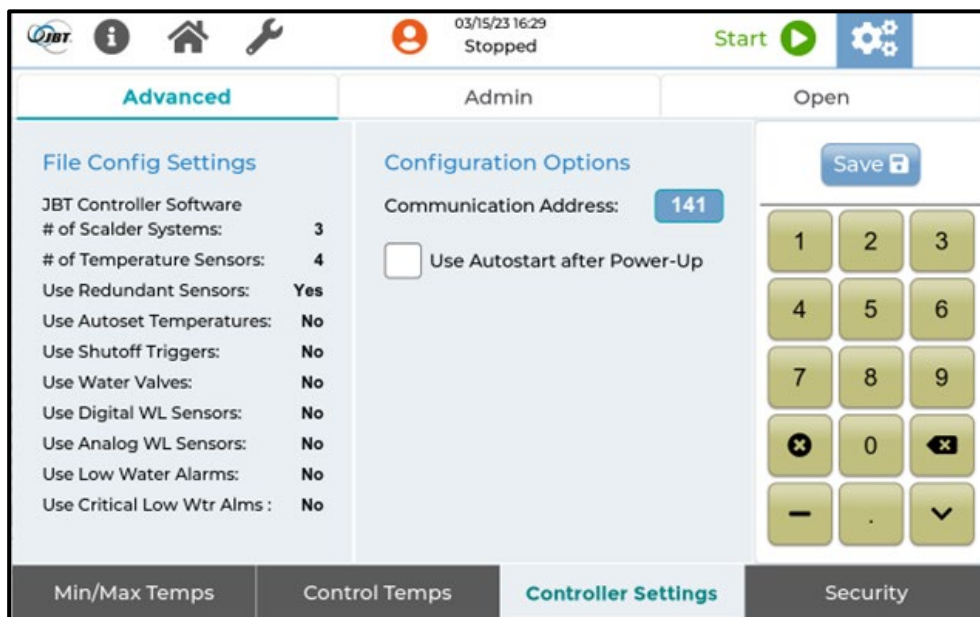
Advanced Settings (Cogs) – Advanced – Controller



Navigation: Adv Settings (Cogs) – Advanced (Upper Tab) – Controller (Lower Tab)

Security Level: Level 3 Advanced Password or above

These values are usually set during the initial installation.

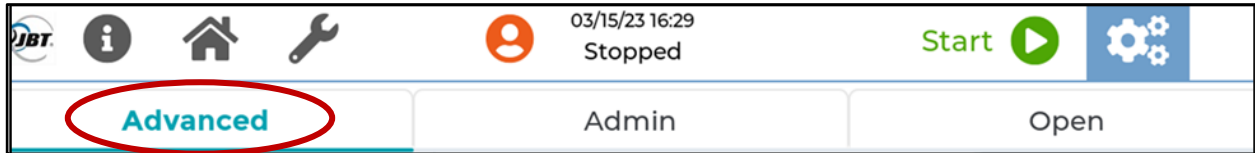


File Config Settings: Configuration settings that are specified in your plant's configuration file. This section is for reference only and provides information about how your system is configured.

Configuration Options

Use Autostart after Power-Up: When checked, the system will automatically start 20 seconds after power-up. If unchecked, the system will require a manual press of the start button on the header navigation bar.

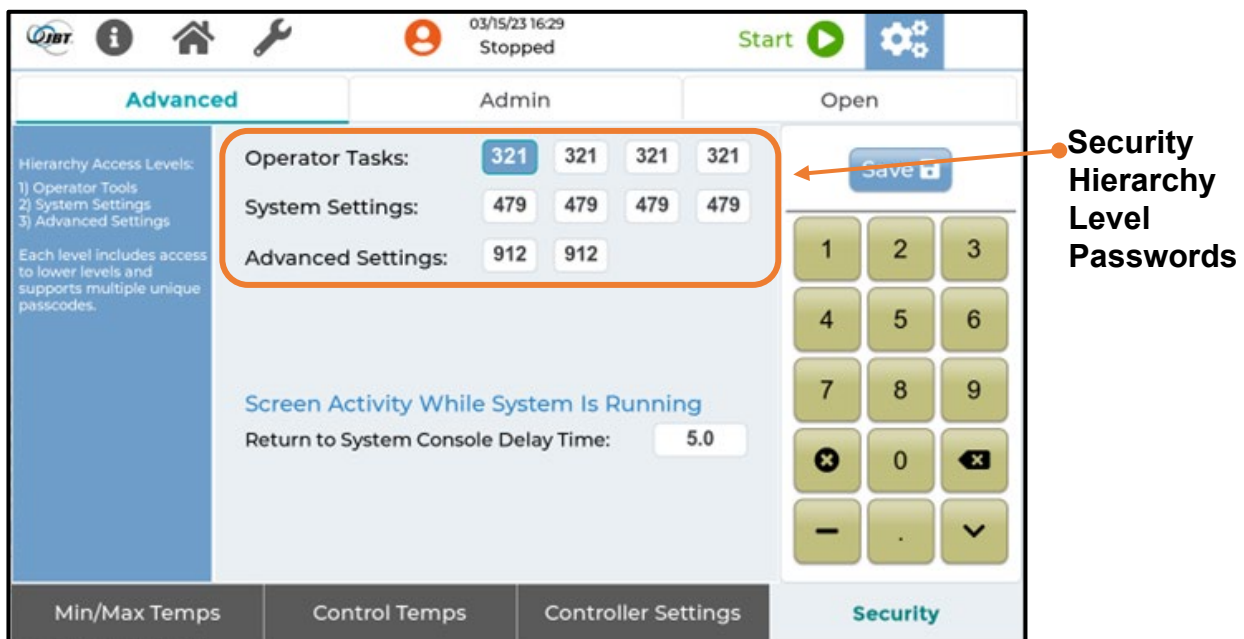
Advanced Settings (Cogs) – Advanced – Security



Navigation: Adv Settings (Cogs) – Advanced (Upper Tab) – Security (Lower Tab)

Security Level: Level 3 Advanced Password

The passcodes are arranged in a **hierarchy style managed by the plant.**



Security Hierarchy

Passwords are evaluated by security level, and higher level password work on lower level tasks. For example, an Operator password is only valid on Operator tasks, were as System level password work on both Operator and System levels, and Advanced password work on Operator, System and Advanced tasks.

Level 1 Operator password are required for Operator type activity such as enabling reject buttons.

Level 2 System Setting password are required for system configuration changes. These password should only be given to those who have the knowledge and authority to change system settings. Level 2 password also work for Level 1 Tasks.

Level 3 Advanced Setting password are required for advanced system settings that should only be done by trained individuals. Level 3 password also work for lower levels.

OPERATIONS AND INSTRUCTIONS

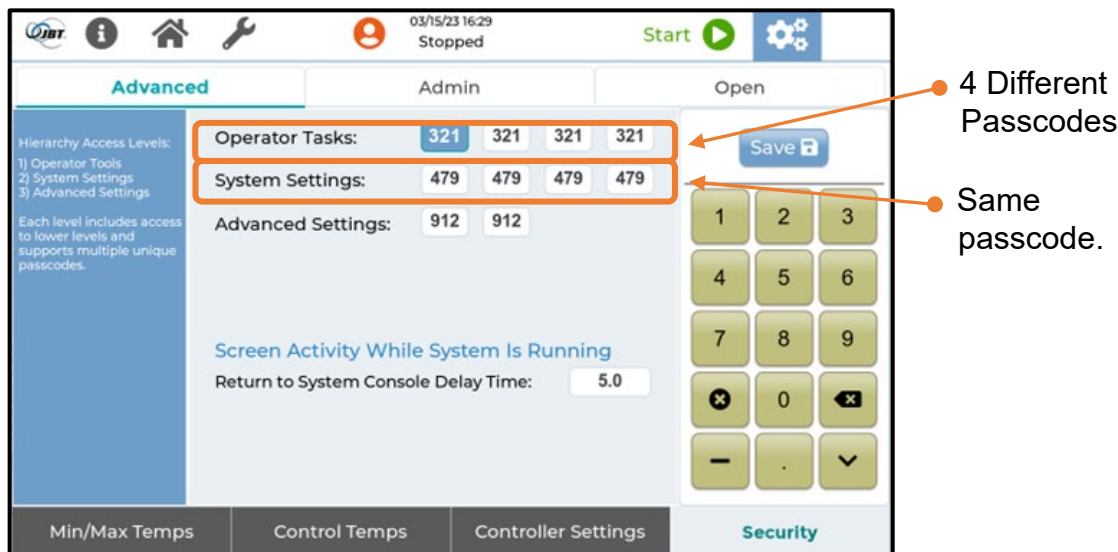
ADVANCED SETTINGS - SECURIT

Return to System Console Delay Time

If the system is running, the home screen/system console is not the active screen, and there has been no user activity, the system will return to the home screen after the configured amount of time has passed.

Plant Managed Passwords

Gen 2 provides a designated login to manage passwords at the plant level. The Security Admin password only works on the Security Screen and can't be used to perform system tasks. Operator and System levels support up to 4 different passwords, and Advanced Level support 2 passwords.



Invalid Passwords

Passwords must be three digits and can not start with 0. An invalid password entry will be changed back to the original value and a red border added.

When to Assign 4 Different Passwords

The Operator Tasks example above shows four different passwords which can be given to the various operators as needed.

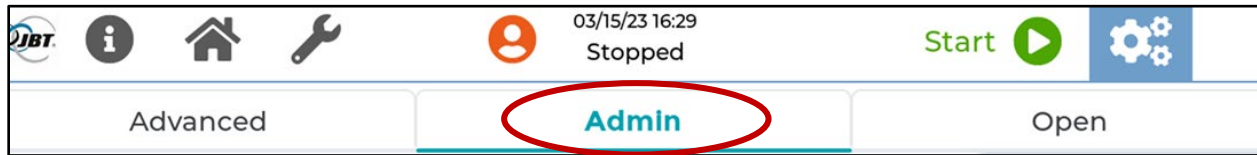
When to Assign Same Passwords

For plants transitioning from Gen1 to Gen2, it's possible that only one or two people need a system level password, or everyone is used to using a certain password. **Using the same password allows the plant to continue with their Gen1 password and assign other passwords at their convenience.**

OPERATIONS AND INSTRUCTIONS

ADVANCED SETTINGS - GENERAL

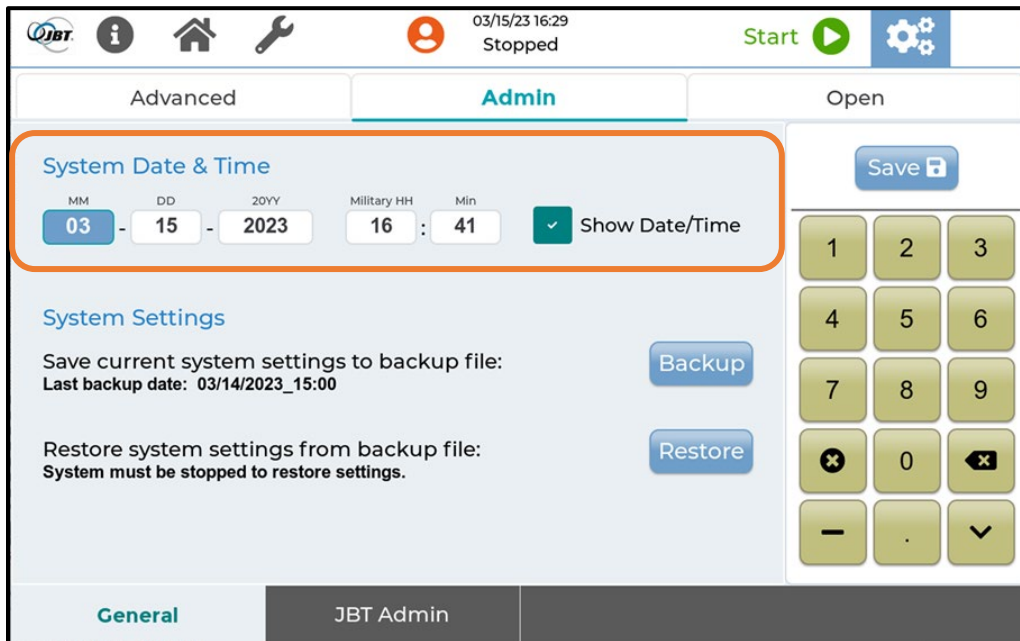
Advanced Settings - Admin - General



Navigation: Adv Settings (Cogs) – Admin (Upper Tab) – General (Lower Tab)

Security Level: Level 3 Advanced Password or above

The General Admin screen contains settings and tools that are unrelated to your system's configuration or operation.



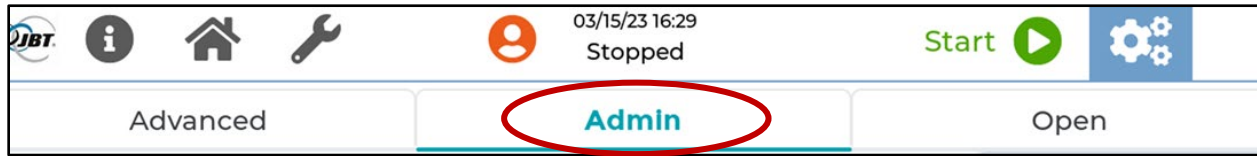
System Date & Time: The JBT controller has a battery-backed up clock for maintaining the correct Date and Time when the controller is not connected to the internet. Enter the correct Date and Time after logging in with your password. Then press the Save button.

Exit to Operating System: Stop the system and exit the JBT software to the Operating System.

OPERATIONS AND INSTRUCTIONS

ADVANCED SETTINGS - JBT ADVANCED ADMIN

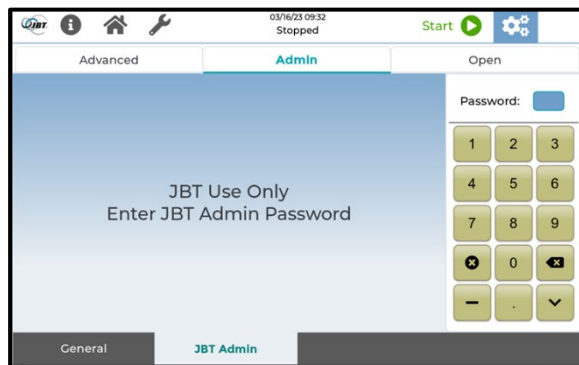
Advanced Settings - Admin – JBT Adv Admin



Navigation: Adv Settings (Cogs) – Admin (Upper Tab) – JBT Adv Admin (Lower Tab)

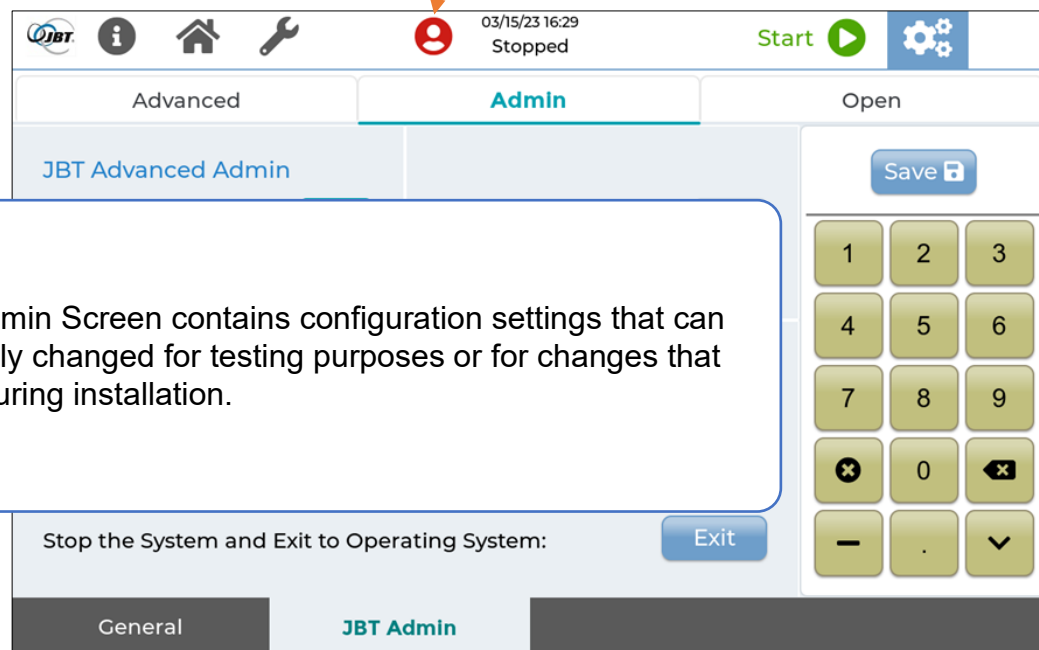
Security Level: JBT Super User (JBT Use Only)

The JBT Admin screen contains settings and tools that are used exclusively by a JBT Technician during installation.



JBT Admin is logged in and passwords are disabled.

Touch icon to logout as JBT Admin.

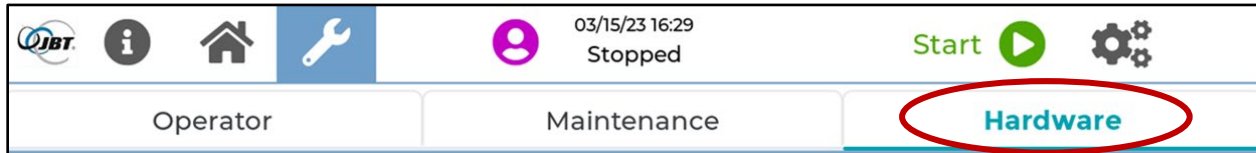


SANITATION

SANITATION PROCEDURES

There are no sanitation procedures for this manual.

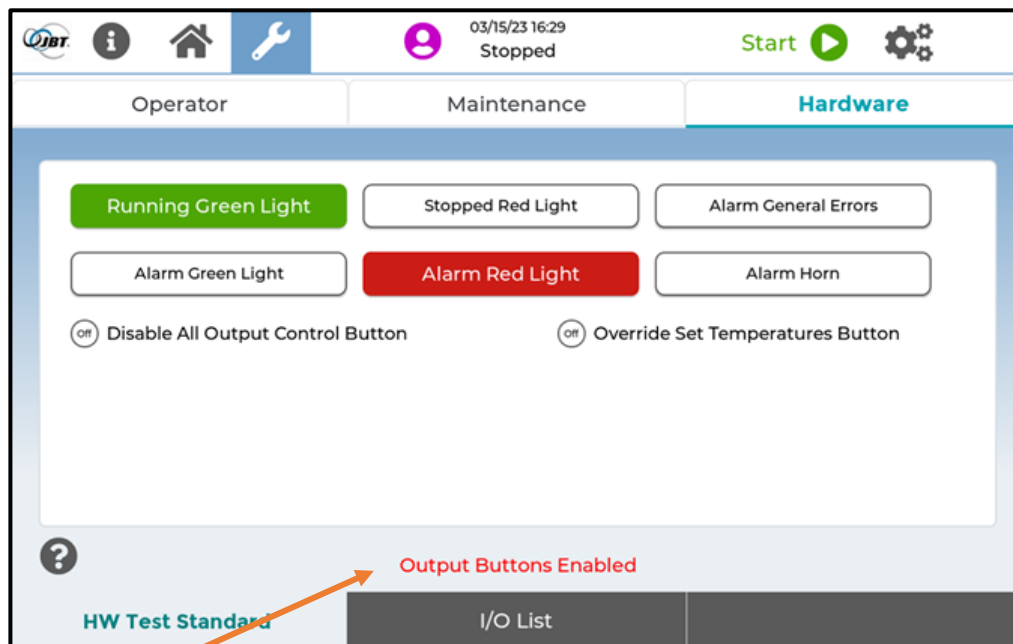
Tools (Wrench) – Hardware – HW Test Standard



Navigation: Tools (Wrench) – Hardware (Upper Tab) – HW Test Standard

Security Level: Level 2 System passcode or above

Your Hardware Test screen will reflect the inputs and outputs for your system.



Status Message

Verify Inputs

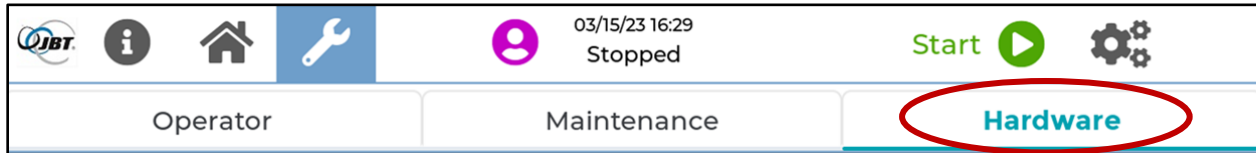
The input buttons are show as small circles which reflect the current state of the input.

Pushbutton States: ☐ Off ☒

Test Outputs

Use the output buttons to test the signal used to control hardware such as gates and external lights. The output buttons also reflect the state of the output while the system is running. Touch the appropriate button to verify the correct response.

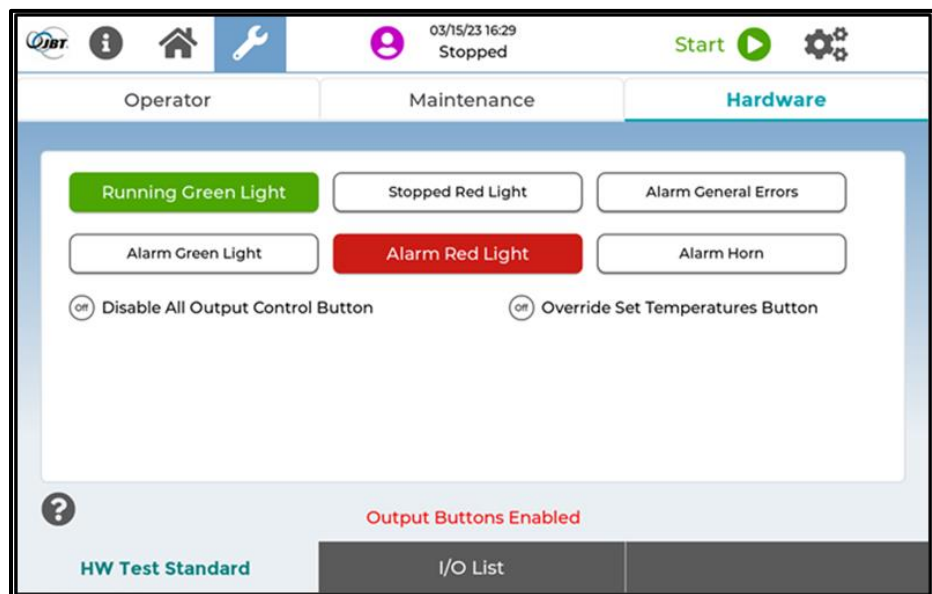
Tools (Wrench) – Hardware – HW Test Standard Optional



Optional Hardware

The JBT Scaldex System supports the use of the following optional hardware.

Optional system status and alarm lights on the hardware test screen.



The following hardware alarm lights/horn are recommended, but installation is based on plant needs. These items will appear on the Hardware Test screen even if the hardware is not installed.

Running Green/Red Light indicates if the system is currently running and **able to control steam valves**. If steam output is disabled for any reason, the red light will be on even if the system is still technically 'running' on the HMI display.

The **General Errors Alarm** is activated when a system hardware error is detected, such as a sensor not responding correctly, or a burned-out signal is detected.

Red/Green Alarm Lights are used to show if temperatures are in or out of a specified temperature range. The **Alarm Horn** is activated at the same time.

MAINTENANCE AND TROUBLESHOOTING

HARDWARE TEST ADDITIONAL

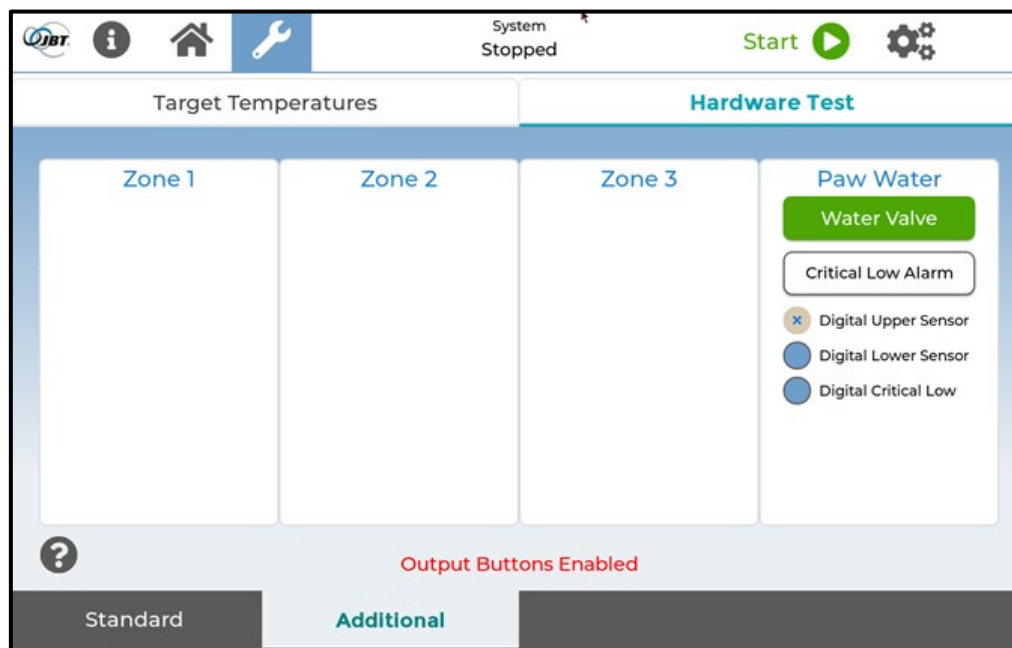
Tools (Wrench) – Hardware – HW Test Additional



Navigation: Tools (Wrench) – Hardware (Upper Tab) – HW Test Additional (Lower)

Security Level: Level 2 System passcode or above

Note: The 'Additional' tab is only visible if optional inputs/outputs are configured that relate only to specific tanks. Shared inputs/outputs are shown on the 'Standard' tab.



In the sample screen shot, the Paw Water tank uses additional inputs/outputs.

Verify Inputs

The input buttons are shown as small circles which reflect the current state of the input. The water level sensors are shown as submerged in water  or above water .

Test Outputs

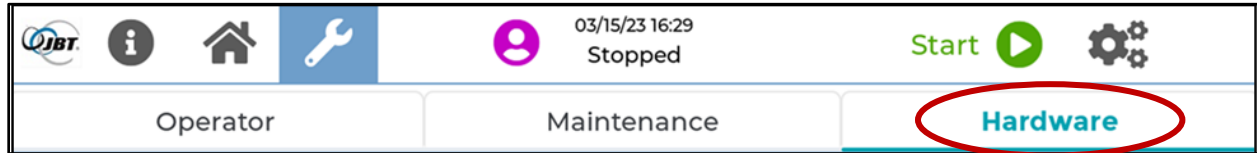
Use the output buttons to test the signal used to control hardware such as external lights, horns, or water valves. The output buttons also reflect the state of the output while the system is running. Touch the appropriate button to verify the correct response.

See Settings (Cogs) – Advanced (Upper Tab)- I/O List (Lower Tab) for I/O assignments.

MAINTENANCE AND TROUBLESHOOTING

I/O LIST

Tools (Wrench) – Hardware – I/O List



Navigation: Tools (Wrench) – Hardware (Upper Tab) – I/O List (Lower Tab)

Security Level: Level 3 Advanced Password or above

The I/O (Inputs/Outputs) screen provides a convenient list of the inputs and outputs configured for your plant's installation. Testing these inputs/outputs is performed on Hardware Test.

The screenshot shows the 'I/O List' screen within the 'Hardware' tab. The interface is divided into four columns: 'Primary Inputs', 'Control Outputs', 'Digital Inputs', and 'Digital Outputs'. The 'Primary Inputs' column lists Tank 1 (41:0), Tank 2 (41:1), and Tank 3 (41:2). The 'Control Outputs' column lists Tank 1 (61:0). The 'Digital Inputs' column lists 'Disable All Control' (1) and 'Override Set Temps' (2). The 'Digital Outputs' column lists 'Running Green Light' (21), 'Stopped Red Light' (22), 'Alarm Errors' (23), 'Alarm Green Light' (24), 'Alarm Red Light' (25), and 'Alarm Horn' (26). Below the table, there are two buttons: 'HW Test Standard' and 'I/O List'.

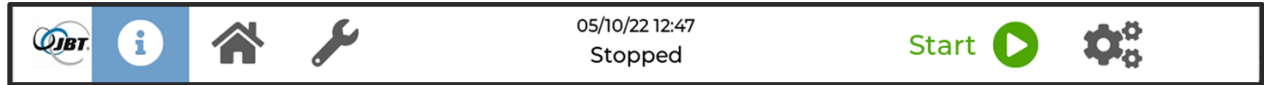
Primary Inputs	Control Outputs	Digital Inputs	Digital Outputs
Tank 1: 41:0	Tank 1: 61:0	Disable All Control: 1	Running Green Light: 21
Tank 2: 41:1		Override Set Temps: 2	Stopped Red Light: 22
Tank 3: 41:2			Alarm Errors: 23
Redundant Inputs			Alarm Green Light: 24
Tank 1: 42:0			Alarm Red Light: 25
			Alarm Horn: 26

Your I/O List will be different than the sample shown here.

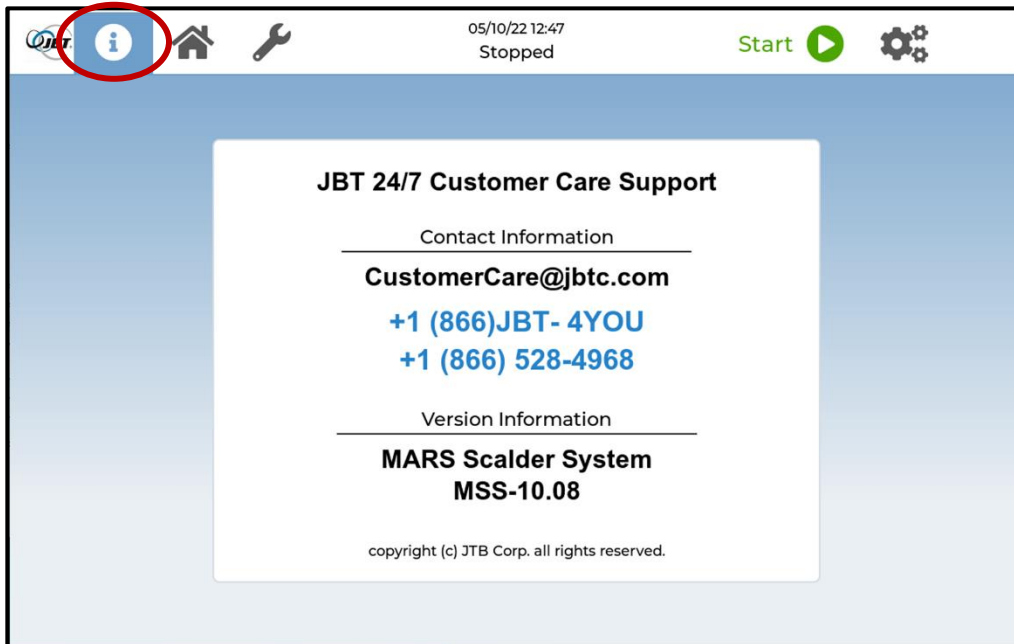
MAINTENANCE AND TROUBLESHOOTING

INFO AND ABOUT/TROUBLESHOOTING

Info and About



Company, Contact, and Software Version Information



Contact Info

Phone: 1 - 866 – JBT – 4YOU
1 - 866 – 528 - 4968
Email: CustomerCare@jbt.com

Contact a JBT Service Technician

If you have a question about your **JBT System** or require service after trying to troubleshoot the issue, please contact us using the contact information provided on your system's Info screen.

Troubleshooting

The **Hardware Test** screen is a good place to start when experiencing problems with the correct operation of your **JBT System**.

If all the external hardware appears to be functioning properly on the **Hardware Test** screen, then diagnosing the problem may require assistance from a JBT service technician.

ASSEMBLY DRAWINGS

The following pages include the Bill of Materials and Assembly Drawings for SN4448.



Bill of Materials, Single Level

Page: 1

ReportName:50056

JBT Marel Corporation

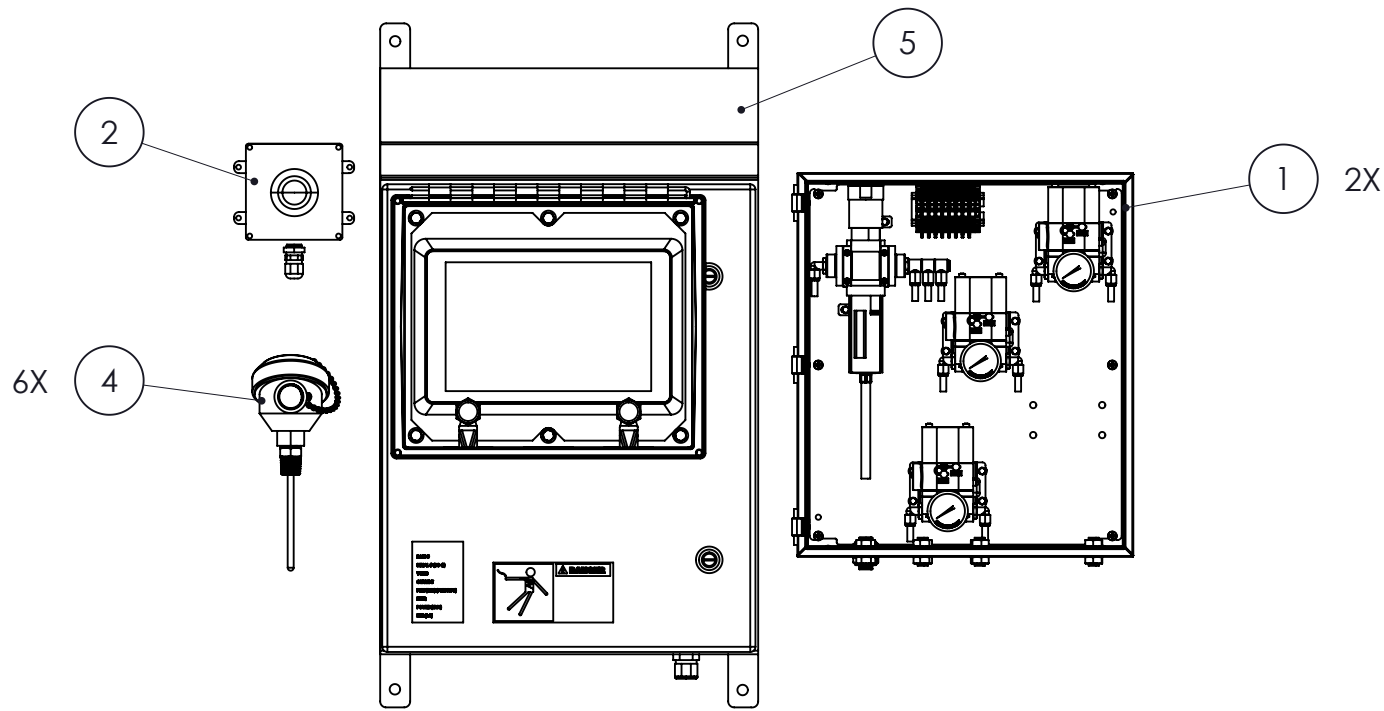
2001 Courtright Rd., Columbus, OH 43232
P: (614) 253-8590 F: (614) 253-6966
www.jbtc.com

September 19, 2025

Item No:
SN4448

MARSFPS-SCLDR1, SCALDER CONTROL, 1 LINE

<u>Item No.</u>	<u>Description</u>	<u>Quantity Per</u>	<u>UOM</u>
MRS1261-SS	ENCL ASSY, TRANSDUCER, 3, SS	2	EA
MRS1286	BTN ASSY,ILLUM,COMMITTED	1	EA
MRS4003	RTD Wire (6 conductor)	5,000	FT
MRS5015	4in KPP Polyhead Dual RTD	6	EA
MSCLDS-10004	SCALDER SYST, JBT, 6 INPUT, 6 OUTPUT	1	EA



NOT SHOWN: 3

5	1	MSCLDS-10004	SCALDER SYST, JBT, 6 INPUT, 6 OUTPUT
4	6	MRS5015	4in KPP Polyhead Dual RTD
3	5000	MRS4003	RTD WIRE (6 CONDUCTOR)
2	1	MRS1286	BTN ASSY,ILLUM,COMMITTED
1	2	MRS1261-SS	ENCL ASSY, TRANSDUCER, 3, SS
ITEM NO.	QTY.	PART NUMBER	DESCRIPTION

UPDATED DESCRIPTION		Description of Change	
Revision	ECN #	By	Date
Current Revision Information			
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THIRD ANGLE PROJECTION

UNLESS OTHERWISE SPECIFIED:
ALL DIMENSIONS ARE IN INCHES
TOLERANCES

X.X ± 0.1 FRACT ±1/16
X.XX ± 0.01 ANGLE ±0.25°
X.XXX ± 0.005

SURFACE FINISH 250 MICROINCH.
BREAK ALL SHARP EDGES 0.015.
CONCENTRICITY 0.01 TIR.
WELD BEAD 1/8" TYP

LOCATIONS

Columbus, OH. USA (614) 253-8590
Gainesville, GA. USA (770) 532-4766
Chapeco SC Brazil 55 (49) 3328-3322

Material: SEE BOM/DETAILS
Finish: SEE BOM/DETAILS
Heat Treat: SEE BOM/DETAILS
Designed By: martist 09/19/2025
Detailed By: martist 09/19/2025
Approved By: mironeg 09/23/2025

PRIME EQUIPMENT GROUP

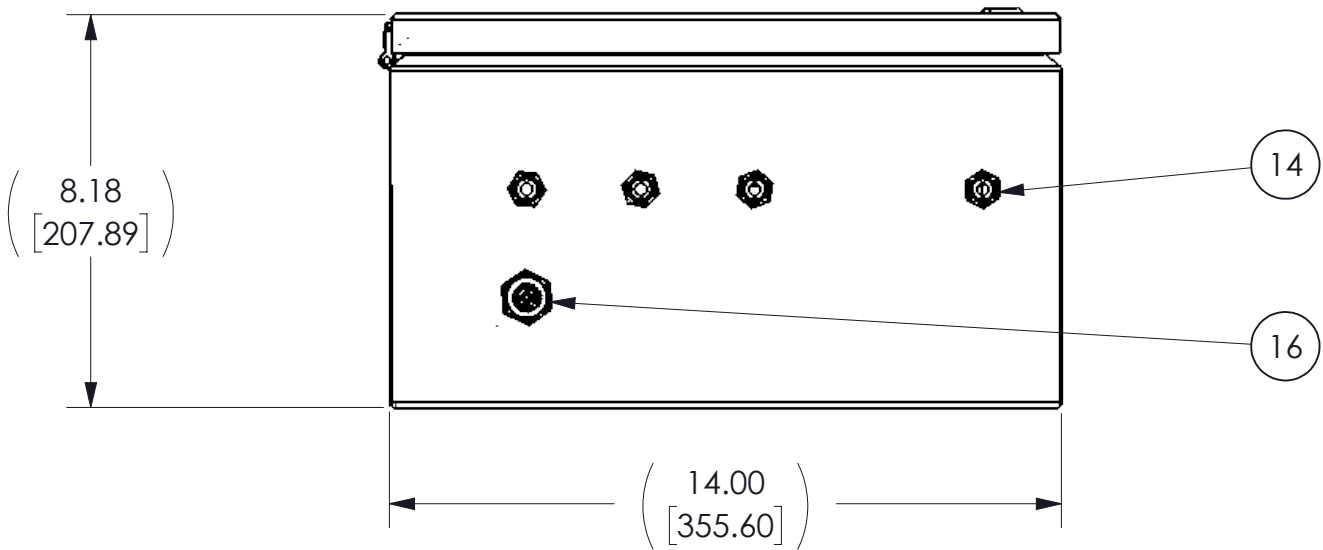
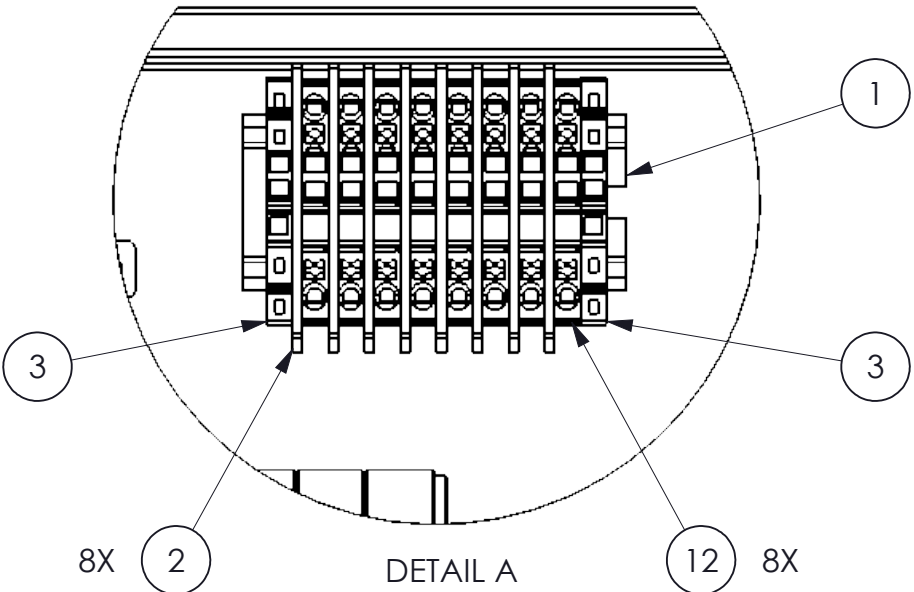
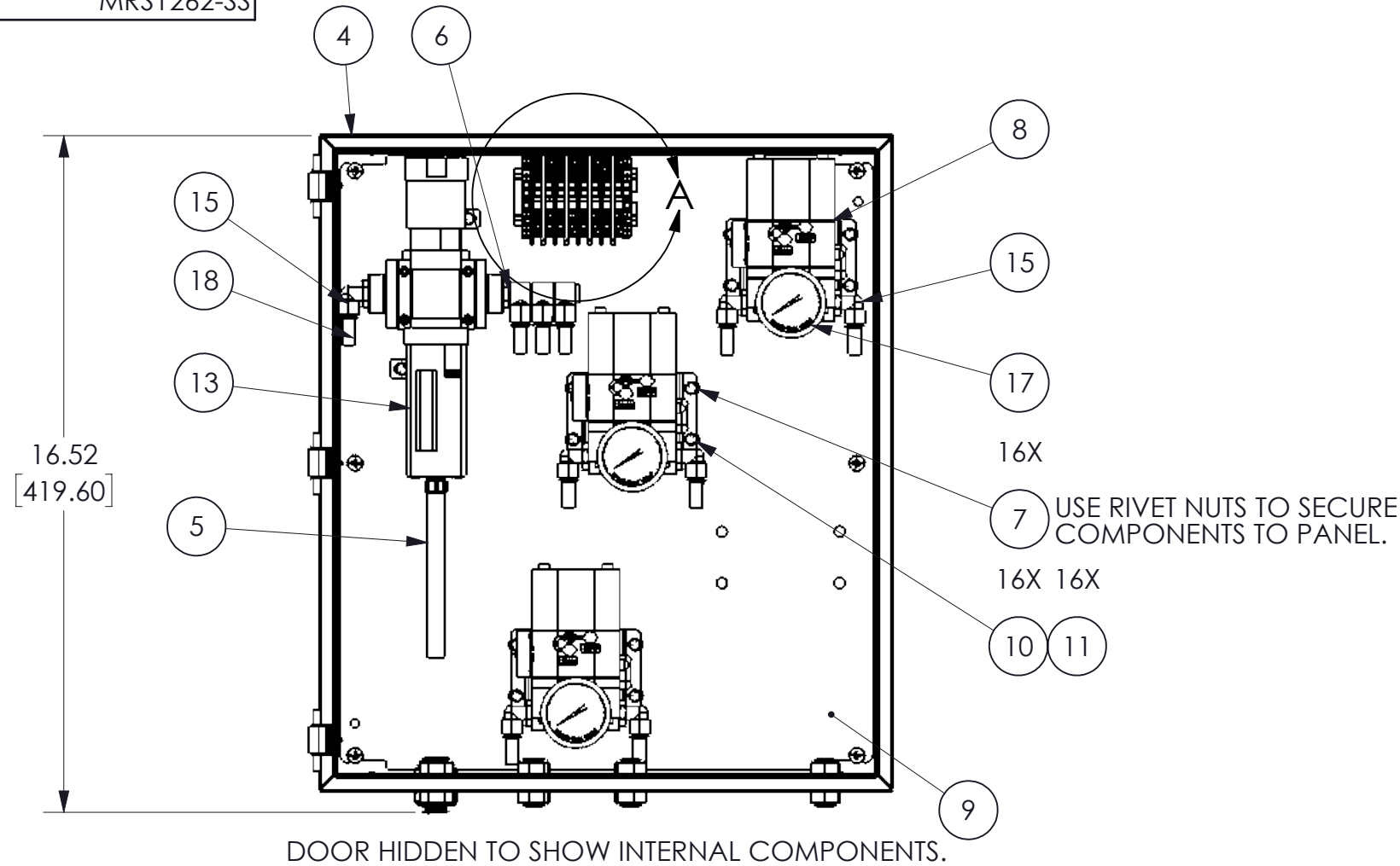
MARSFPS-SCLDR1, SCALDER CONTROL, 1 LINE

Size Part Number

A SN4448

Rev. A

SHEET 1 OF 1



NOTE:
ALL DIMENSIONS ARE IN INCHES, MILLIMETERS ARE SPECIFIED AS A REFERENCE.

18	5	MRS7511	1/4in Blue Tubing	
17	3	MRS7489	Air Gauge (1/4in) (30 PSI)	
16	1	MRS7469	3/8in Bulkhead Fitting	
15	7	MRS7432	1/4inx1/4in L Air Fitting	
14	4	MRS7429	1/4in Bulkhead Fitting	
13	1	MRS1276	Tsunami Filter-Regulator Combo	
12	8	MRS1222	TERMINAL,26-12AWG,ORA,5.2MM	
11	16	LWSS08	SLW,#8,SS	
10	16	HCSS008C0050	SCR,CAP,HEXHD,#8-32X1/2,SS	
9	1	84231	BACKPANEL, 16X14, XDCR BOX, SS	
8	3	79136	TRANSDUCER,500-AD,3-27PSIG	
7	16	59209	RIVET, NUT, 8-32, AL	
6	1	50582	FTG,MANF,3OUT,1/4TBG,1/4MNPT	
5	12	43629	TBG,0.38OD,0.25ID,PUR,CLEAR	
4	1	29313	ENCL, 16X14X08, FLTTOP, NEMA4X, SS	
3	2	27500	END CLAMP, TERM, GRAY, 5.15MM	
2	8	27499	PART PLT, TERM, 2MM, GREY, F/27498	
1	1	11874	DIN RAIL, STEEL, 35MM X 15MM	3
ITEM NO.	QTY.	PART NUMBER	DESCRIPTION	LENGTH

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TOLERANCES
X.X ± 0.1 FRACT ± 1/16
X.XX ± 0.01 ANGLE ± 0.25°
X.XXX ± 0.005
SURFACE FINISH 250 MICROINCH.
BREAK ALL SHARP EDGES 0.015.
CONCENTRICITY 0.01 TIR.
WELD BEAD 1/8" TYP

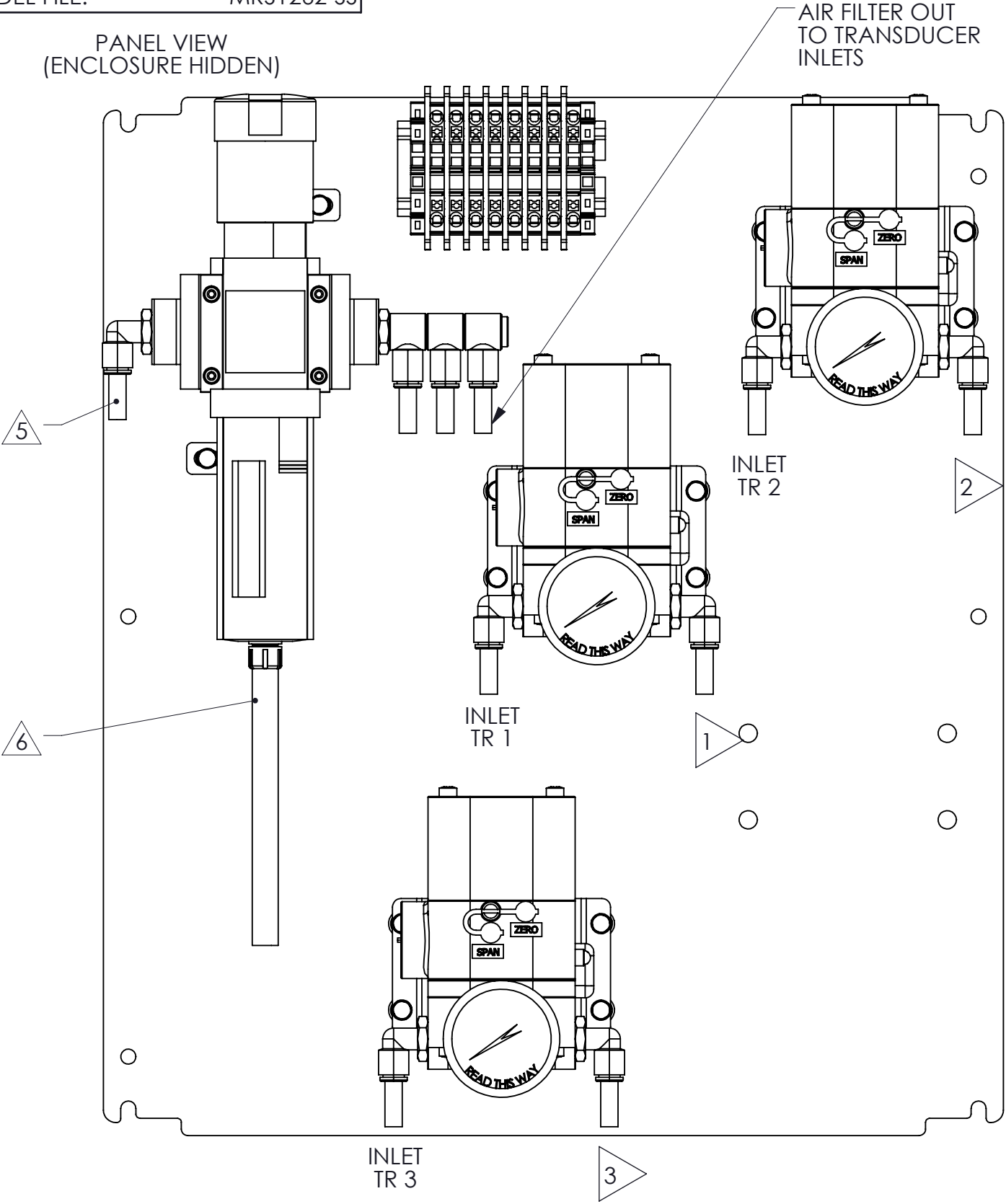
LOCATIONS	
Columbus, OH. USA	(614) 253-8590
Gainesville, GA. USA	(770) 532-4766
Chapeco SC Brazil	55 (49) 3328-3322
Material:	SEE BOM/DETAILS
Finish:	SEE BOM/DETAILS
Heat Treat:	SEE BOM/DETAILS
Designed By:	kankrig 11/23/2022
Detailed By:	loganca 02/05/2024
Approved By:	mironeg 02/08/2024

PRIME EQUIPMENT GROUP

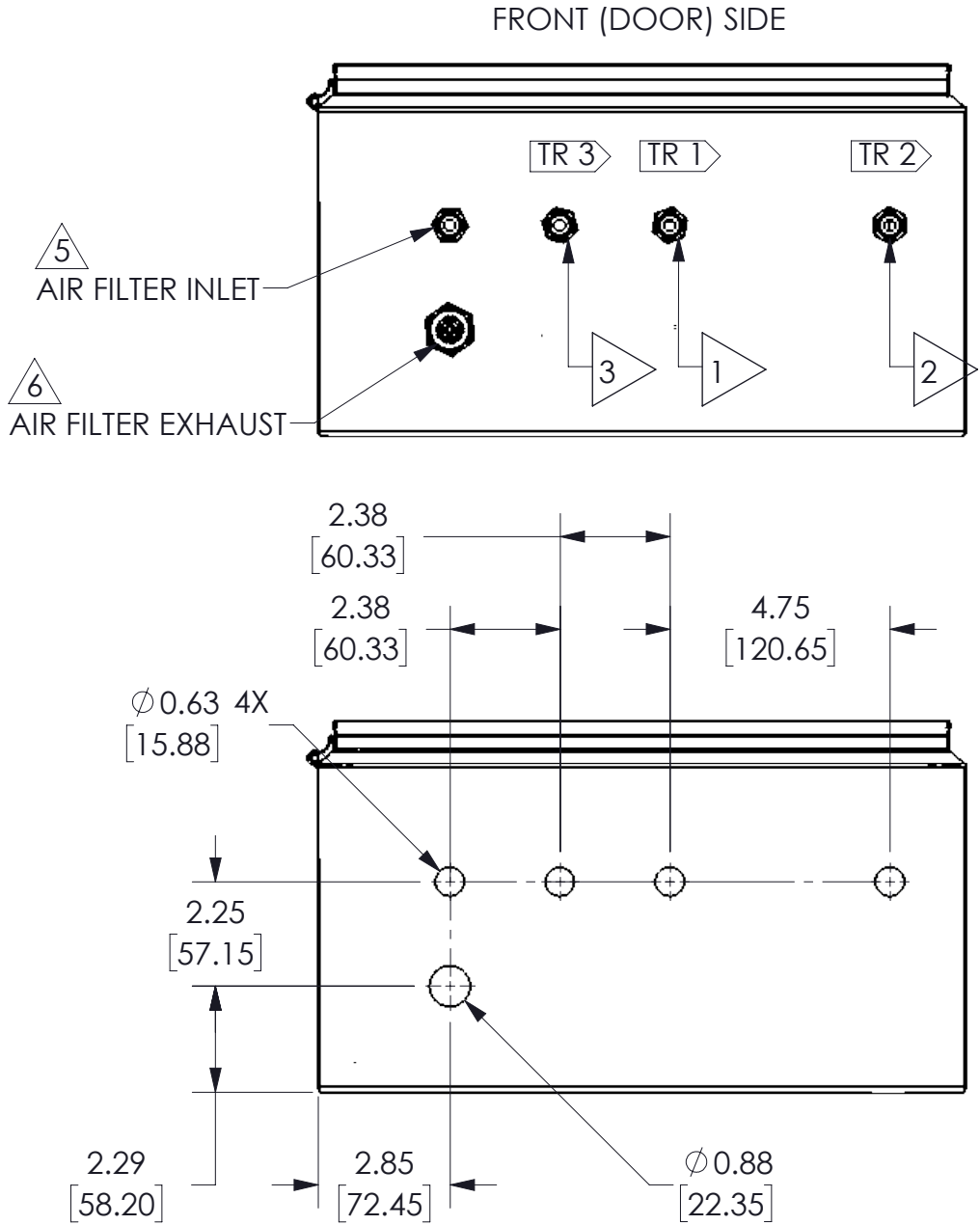
ENCL ASSY, TRANSDUCER, 3, SS

Size	Part Number	Rev.
B	MRS1261-SS	A

SHEET 1 OF 2



- NOTES:**
1. LABEL WIRES, TERMINALS, TRANSDUCERS AND FITTINGS AS "TR 1", "TR 2", AND "TR 3" APPROPRIATELY.
 2. LABEL THE AIR FILTER INLET BULKHEAD FITTING AS "INLET" AND EXHAUST FITTING AS "EXHAUST."
 3. USE TEMPLATE "TR-BOX" TO LOCATE AND CUT ALL HOLES FOR BULKHEAD FITTINGS.



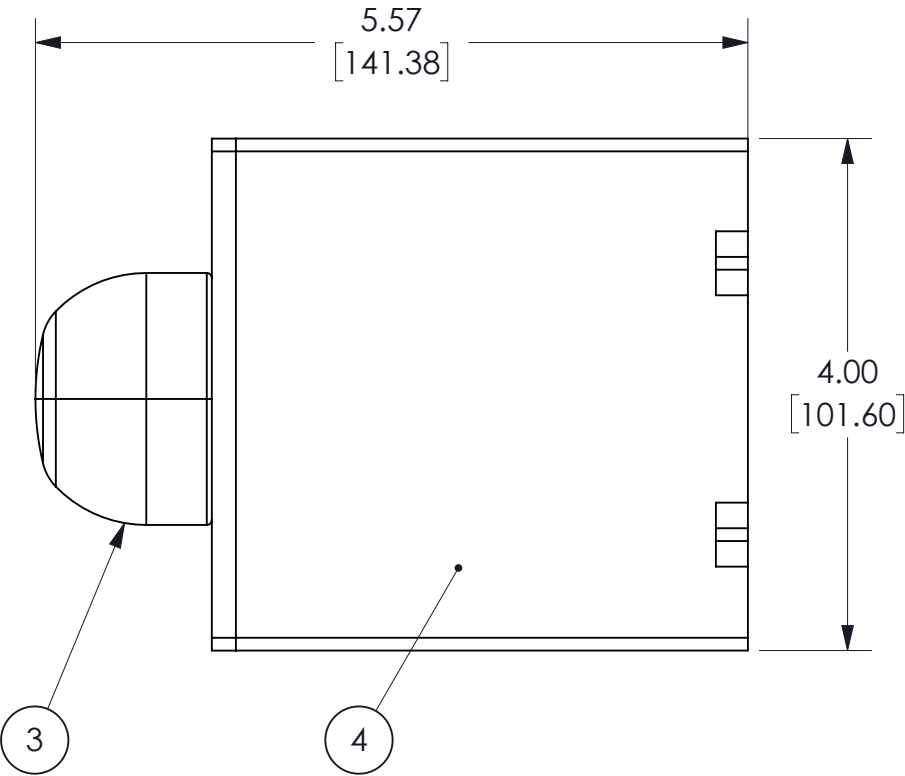
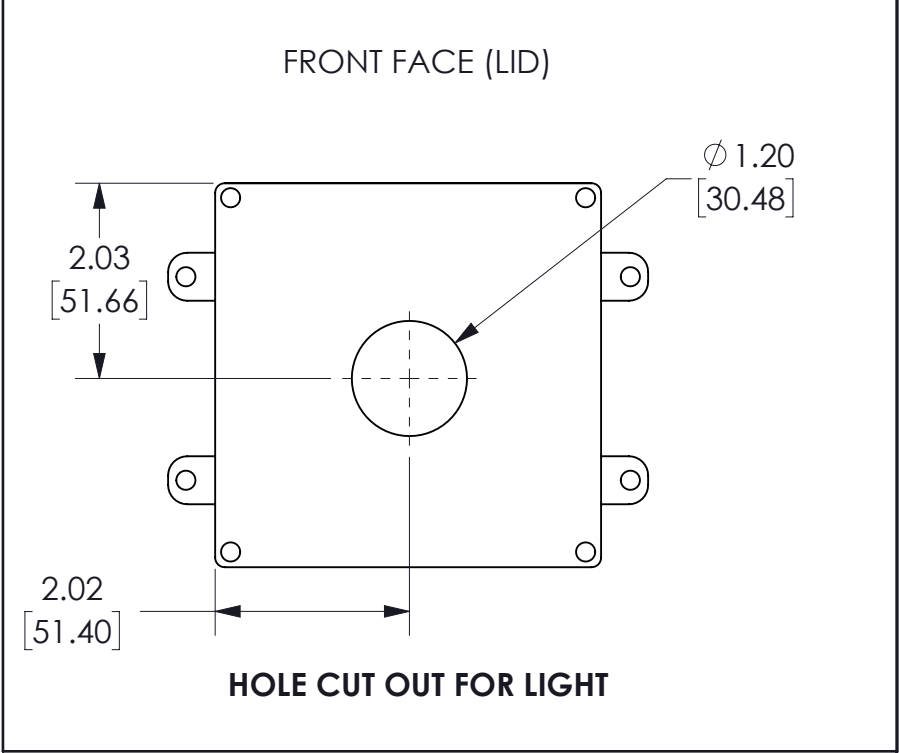
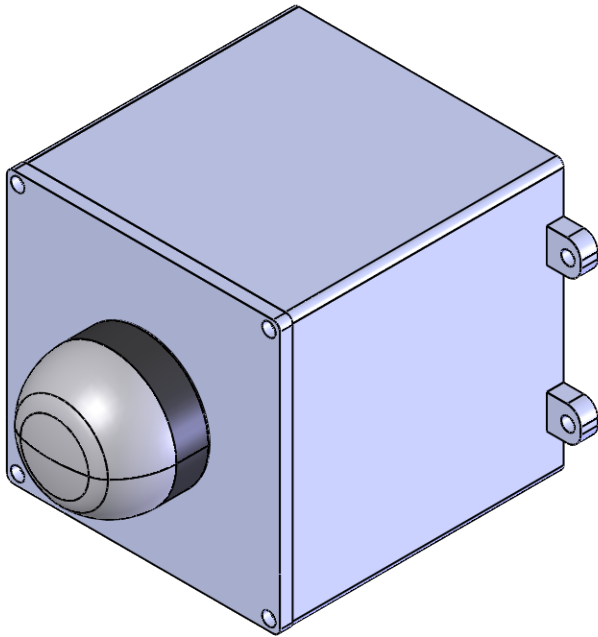
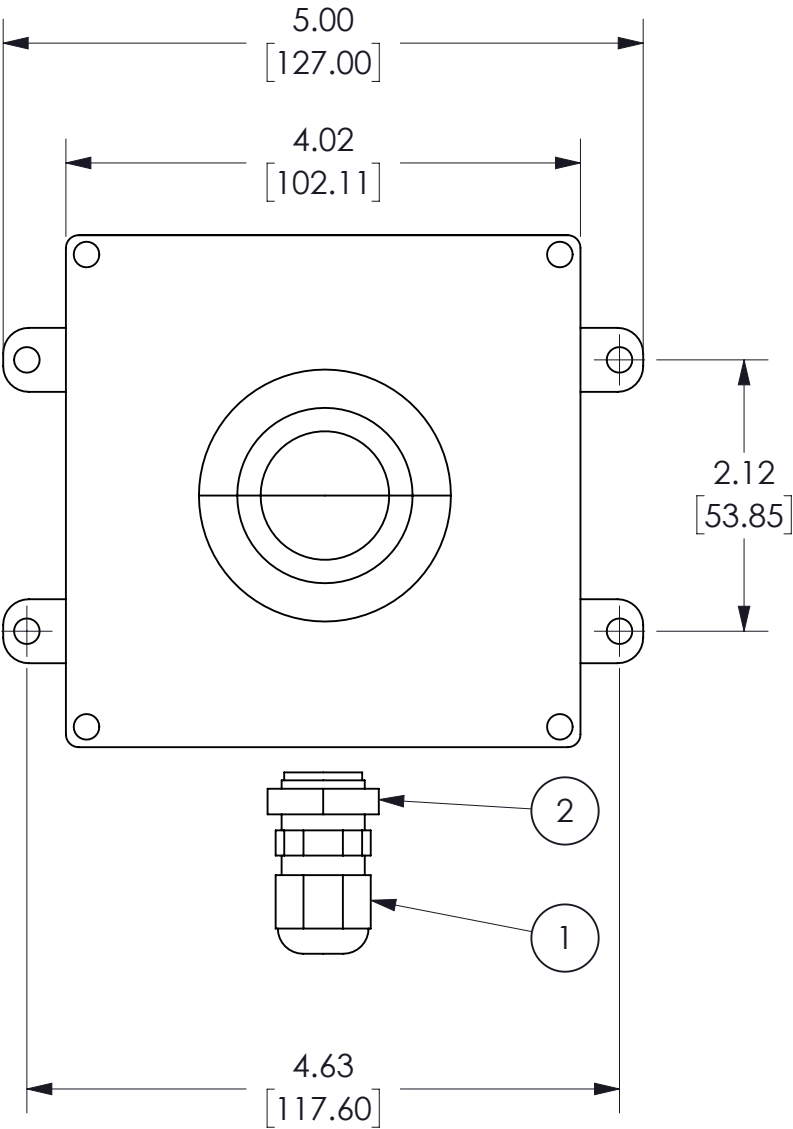
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THIRD ANGLE PROJECTION	
UNLESS OTHERWISE SPECIFIED: ALL DIMENSIONS ARE IN INCHES TOLERANCES	
X.X ± 0.1	FRACT ± 1/16
X.XX ± 0.01	ANGLE ± 0.25°
X.XXX ± 0.005	
SURFACE FINISH 250 MICROINCH. BREAK ALL SHARP EDGES 0.015. CONCENTRICITY 0.01 TIR. WELD BEAD 1/8" TYP	

LOCATIONS	
Columbus, OH. USA	(614) 253-8590
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Chapeco SC Brazil	55 (49) 3328-3322
Material:	SEE BOM/DETAILS
Finish:	SEE BOM/DETAILS
Heat Treat:	SEE BOM/DETAILS
Designed By:	kankrig 11/23/2022
Detailed By:	loganca 02/05/2024
Approved By:	mironeg 02/08/2024

PRIME EQUIPMENT GROUP	
ENCL ASSY, TRANSDUCER, 3, SS	
Size	Part Number
B	MRS1261-SS
Rev.	A
SHEET 2 OF 2	



PLACE ITEM 1 AND ITEM 2 INSIDE ENCLOSURE(ITEM 4)

4	1	MRS4201	ENCL, 04X04X04, NEMA4, PLASTIC
3	1	MRS4132	BTN,TOUCH,ILLUM,50MM,NO,12-30V,G-R,2M CABLE
2	1	MRS1561	3/8in Connector Nut
1	1	MRS1560	CORD GRIP, 1HOLE, 3/8NPT, STR
ITEM NO.	QTY.	PART NUMBER	DESCRIPTION

NOTE:
ALL DIMESNIONS ARE IN INCHES, MILLIMETERS ARE SPECIFIED AS A REFERENCE.

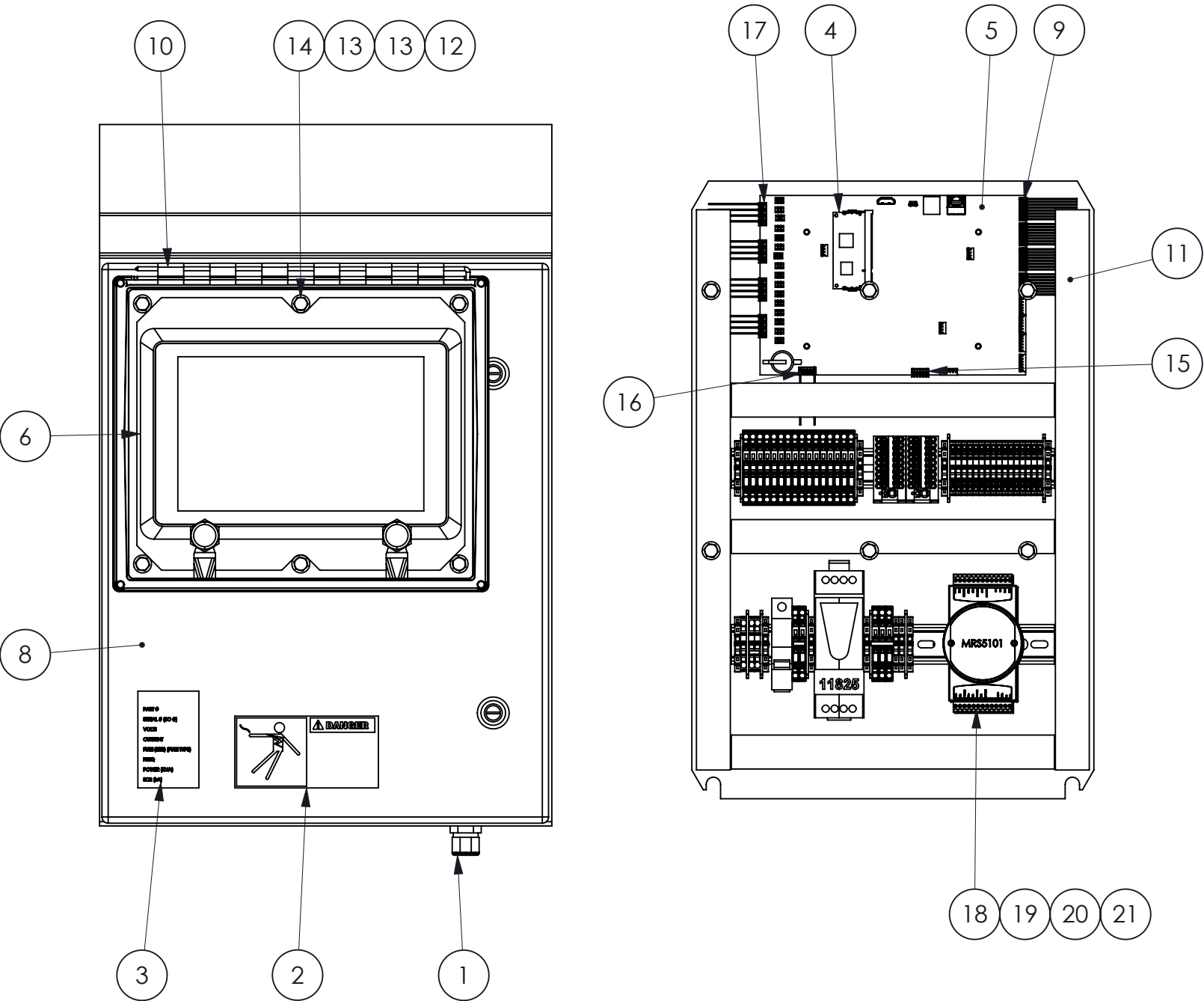
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D	-	mironeg	03/25/2025	UPDATED BOM
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THIRD ANGLE PROJECTION	
UNLESS OTHERWISE SPECIFIED: ALL DIMENSIONS ARE IN INCHES TOLERANCES	
X.X ± 0.1	FRACT ± 1/16
X.XX ± 0.01	ANGLE ± 0.25°
X.XXX ± 0.005	
SURFACE FINISH 250 MICROINCH. BREAK ALL SHARP EDGES 0.015. CONCENTRICITY 0.01 TIR. WELD BEAD 1/8" TYP	

LOCATIONS	
Columbus, OH, USA	(614) 253-8590
Gainsville, GA, USA	(770) 532-4766
Chapeco SC Brazil	55 (49) 3328-3322
Material:	SEE BOM/DETAILS
Finish:	SEE BOM/DETAILS
Heat Treat:	SEE BOM/DETAILS
Designed By:	kankrig 12/21/2022
Detailed By:	kankrig 12/21/2022
Approved By:	mironeg 03/25/2025

PRIME EQUIPMENT GROUP	
BTN ASSY,ILLUM,COMMITTED	
Size	Part Number
B	MRS1286
Rev.	D
SHEET 1 OF 1	



NOT SHOWN: 7

21	1	MRS5102.42	MODULE,INPUT,TEMP,CHNL 42,PRGMD
20	1	MRS5102.41	MODULE,INPUT,TEMP,CHNL 41,PRGMD
19	1	MRS5101.62	MODULE,OUTPUT,TEMP,CHNL 62,PRGMD
18	1	MRS5101.61	MODULE,OUTPUT,TEMP,CHNL 61,PRGMD
17	4	MRS1043	CBL, OUTPUT WIRES W/CONNECTOR
16	1	MRS1042	CBL,POWER WIRE w/CONNECTOR
15	1	MRS1026	CONN, BOARD, 6 PIN
14	6	HCSS025C0100	SCR,CAP,HEXHD,1/4-20X1,SS
13	12	FWUS025	FLW,1/4,SS
12	6	ESNS025C	LKNT,INSR,NYL,1/4-20,SS
11	1	EA120-100001	MARS/JBT CNTRL PANEL ASM, BASIC
10	1	81636	CVR,HMI,12X10,MOD,PLASTIC
9	4	81250	CBL, INPUT WIRE W/CONNECTOR JBT CONTROLLER
8	1	81105	MACH, ENCL, 20X16X08, SLPTOP, JBT SYS
7	1	80787	CBL, HDMI, 3FT
6	1	80742	DISPLAY, 10", PNL MT, ALUM BEZEL, IP65
5	1	80001	BOARD, CNTRL, JBT
4	1	79453-16-CP240-1007	RASPBERRY PI COMPUTE MODULE 3+ / 16GB
3	1	28814	PLT,LEGND,PVC,0,5MM THCK,US-EMLP 85,6X54
2	1	19883	SAFETY STICKER,WARN,VOLTAGE,HORIZONTAL
1	1	10526	CORD GRIP, 0.20-0.35, 1/2NPT, 1HOLE, STR
ITEM NO.	QTY.	PART NUMBER	DESCRIPTION

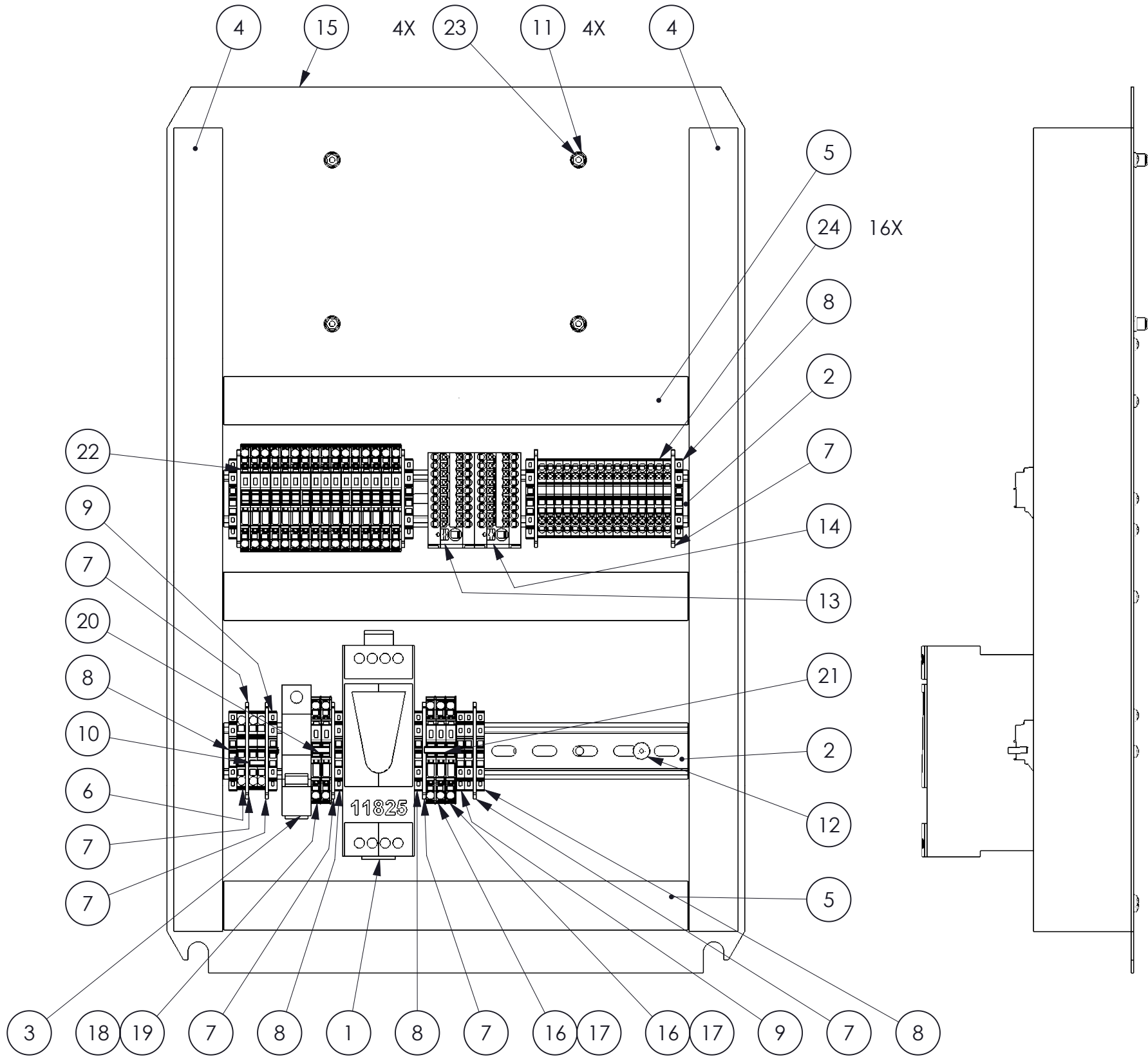
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A	-	-	-	NEW CONFIGURATION
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TOLERANCES
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X.XX ± 0.01 ANGLE ± 0.25°
X.XXX ± 0.005
SURFACE FINISH 250 MICROINCH.
BREAK ALL SHARP EDGES 0.015.
CONCENTRICITY 0.01 TIR.
WELD BEAD 1/8" TYP

LOCATIONS	
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Chapeco SC Brazil	55 (49) 3328-3322
Material:	SEE BOM/DETAILS
Finish:	SEE BOM/DETAILS
Heat Treat:	SEE BOM/DETAILS
Designed By:	martist 11/06/2024
Detailed By:	martist 11/06/2024
Approved By:	mironeg 11/07/2024

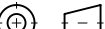
PRIME EQUIPMENT GROUP		
SCALDER SYST, JBT, 6 INPUT, 6 OUTPUT		
Size	Part Number	Rev.
B	MSCLDS-10004	A
SHEET 1 OF 1		



24	16	MRS1223	TERMINAL,26-12AWG,YEL,5.2MM	
23	4	MRS1021	Control Panel Long Stand-Offs	
22	16	84016	FUSE, 315mA, MINI	
21	1	82850	JUMPER, 3P, 6.2MM TERM, GRA	
20	1	82849	JUMPER, 2P, GRAY	
19	2	80044	TERM, HLDR, FUSE, MINI, 120V	
18	2	80038	FUSE, 5A, MINI	
17	3	80037	FUSE, 2A, MINI	
16	19	80036	TERM, HLDR, FUSE, MINI, 24V	
15	1	79926	PNL,BACK,16X20,SS,MARS 16 I/O	
14	1	69567	TERMINAL, DISTR, 19PTS, BK	
13	1	69566	TERMINAL, DISTR, 19PTS, RED	
12	25	59210	RIVET, POP, 3/16, AL	
11	4	59209	RIVET, NUT, 8-32, AL	
10	1	27503	JUMPER,BUS,2P,F/5.2MM TERM,RED	
9	3	27501	TERMINAL, GND, 26-12AWG, GRN/YEL, 5.2MM	
8	8	27500	END CLAMP, TERM, GRAY, 5.15MM	
7	9	27499	PART PLT, TERM, 2MM, GREY, F/27498	
6	3	27498	TERMINAL, 26-12AWG, GREY, 5.2MM	
5	3	17413	WIRE DUCT, 30X60MM, NAR SLOT, GRAY	11.25
4	2	17413	WIRE DUCT, 30X60MM, NAR SLOT, GRAY	19.5
3	1	12974	CIRCUIT BREAKER, 1P, 5A	
2	2	11874	DIN RAIL, STEEL, 35MM X 15MM	11.25
1	1	11825	POWER SUPPLY, 24VDC, 3A, 100-500V	
ITEM NO.	QTY.	PART NUMBER	DESCRIPTION	LENGTH

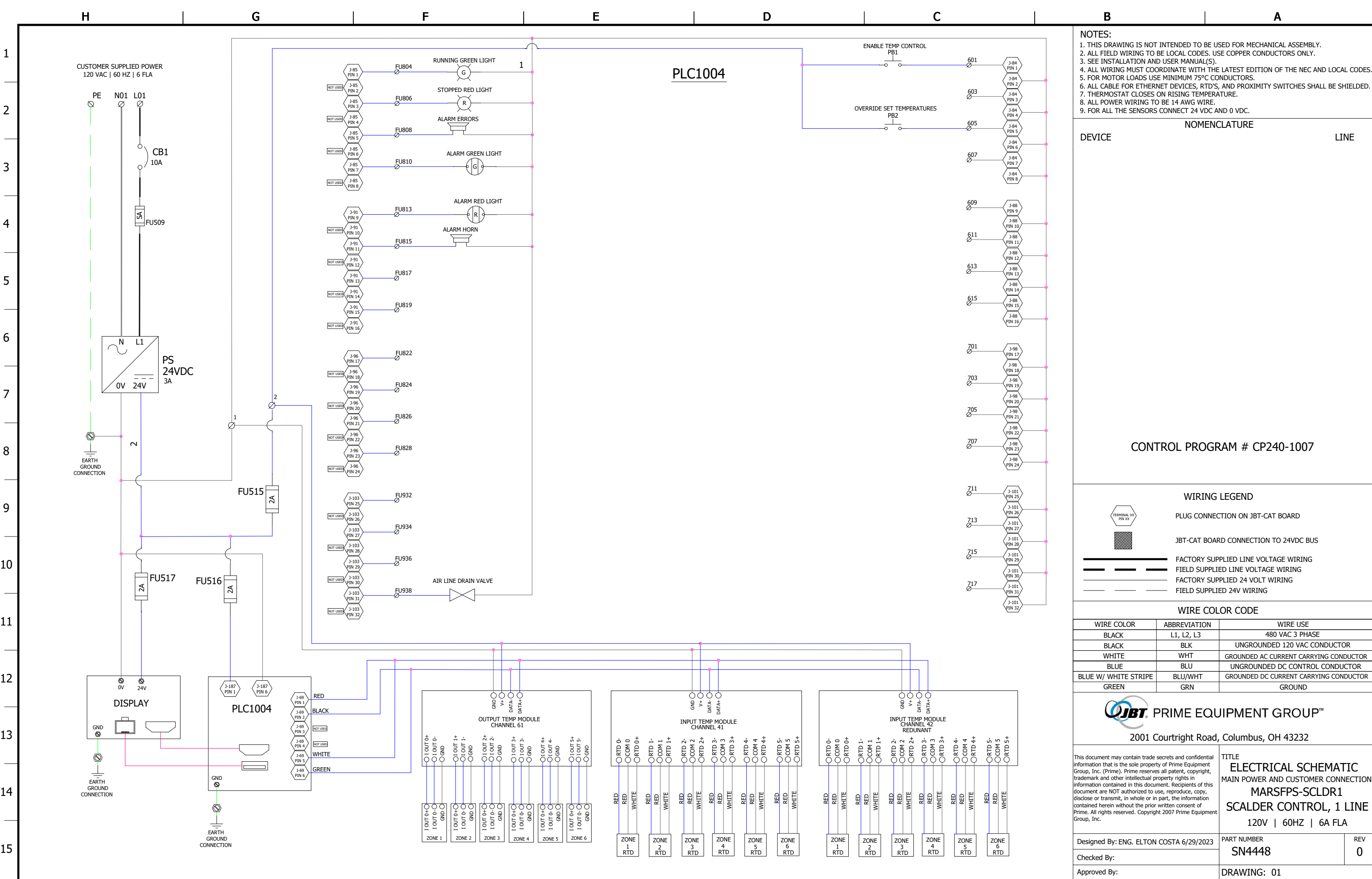
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I	-	martist	06/24/2024	UPDATED BOM
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THIRD ANGLE PROJECTION				LOCATIONS		 PRIME EQUIPMENT GROUP	
UNLESS OTHERWISE SPECIFIED: ALL DIMENSIONS ARE IN INCHES TOLERANCES				Columbus, OH. USA (614) 253-8590		MARS/JBT CNTRL PANEL ASM, BASIC	
				Gainsville, GA. USA (770) 532-4766			
				Chapeco SC Brazil 55 (49) 3328-3322			
X.X ± 0.1 FRACT ± 1/16				Material:			
X.XX ± 0.01 ANGLE ± 0.25°				SEE BOM/DETAILS			
X.XXX ± 0.005				Finish:			
				SEE BOM/DETAILS			
SURFACE FINISH 250 MICROINCH.				Heat Treat:		Size Part Number Rev.	
BREAK ALL SHARP EDGES 0.015.				SEE BOM/DETAILS		B EA120-100001 I	
CONCENTRICITY 0.01 TIR.				Designed By: willswy Jul 22, 21			
WELD BEAD 1/8" TYP				Detailed By: willswy 07/22/2021			
				Approved By: heckert 04/11/2022		SHEET 1 OF 1	

CONTROLS AND ELECTRICAL SCHEMATICS

PART NO.	DESCRIPTION
MSCLDS-10004	SCALDER SYST, JBT, 6 INPUT, 6 OUTPUT



SPARE PARTS LIST

To order spares call 866.528.4968

Email: parts.prime@JBTC.com

Refer to serial number:

Refer to model number: JBT Scalding Control System

Although many parts may seem similar between 2 machines, due to wear and adjustments they become machine specific. You will cause damage to your machine if you swap parts with other machines.

PART NO.	DESCRIPTION	QTY.
11825	POWER SUPPLY, 24VDC, 3A, 100-500V	1
12974	CIRCUIT BREAKER, 1P, 5A	1
79136	TRANSDUCER, PRESS, 3-27PSIG	3
79453-16-CP240-1007	RASPBERRY PI COMPUTE MODULE 3+ / 16GB	1
80001	BOARD, CNTRL, JBT	1
80036	TERMINAL, HLDR, FUSE, MINI, 24V	5
80037	FUSE, 2A, MINI	10
80038	FUSE, 5A, MINI	5
80742	DISPLAY, 10", PNL MT, CAP, ALUM BEZEL, IP65	1
80787	CABLE, HDMI, 3FT	1
81250	CABLE, INPUT WIRE W/CONNECTOR JBT CONTROLLER	4
81636	CVR, HMI, 12X10, MOD, PLASTIC	2
84016	FUSE, 315mA, MINI	16
MRS1042	CABLE, POWER WIRE w/CONNECTOR	1
MRS1043	CABLE, OUTPUT WIRES W/CONNECTOR	4
MRS1276	Tsunami Filter-Regulator Combo	1
MRS1286	BTN ASSY, ILLUM, COMMITTED	1
MRS5013	2in KPP Polyhead Dual RTD	2
MRS5015	4in KPP Polyhead Dual RTD	6
MRS5101.61	MODULE, OUTPUT, TEMP, CHNL 61, PRGMD	1
MRS5101.62	MODULE, OUTPUT, TEMP, CHNL 62, PRGMD	1
MRS5102.41	MODULE, INPUT, TEMP, CHNL 41, PRGMD	1

SPARE PARTS LIST

MRS5102.42	MODULE,INPUT,TEMP,CHNL 42,PRGMD	1
MRS7489	GAUGE,30 PSI,1.50IN,1/4NPT,CB	3

TECHNICAL PARTS INFORMATION

PART NO.	DESCRIPTION
11825	PWRSPLY,100-550VAC,24VDC*

W9148941401104

ABL 8RPS24030

**ABL 8RPS24050/100
ABL 8RPM24200
ABL 8WPS24200/400**

① - 35 mm DIN rail mounting clip.
② - Snap-on label.
③ - Voltage and output current status LED **Uout - Iout**.
④ - Diagnostic Output (Normally Open dry contact 11-14)

① - Klemmfeder auf 35mm-Schiene.
② - Etikettenhalter.
③ - Status-LED Ausgangsspannung und -strom **Uout - Iout**.
④ - Diagnoseausgang (Schließer Trockenkontakt 11-14).

① - Molla di aggancio su profilato 35 mm.
② - Contrassegno agganciabile.
③ - LED di stato della tensione e della corrente di uscita **Uout - Iout**.
④ - Uscita diagnostica (contatto a secco Normalmente Aperto 11-14).

① - Ressort de clipsage sur profilé 35 mm.
② - Repère encliquetable.
③ - LED d'état de la tension et du courant de sortie **Uout - Iout**.
④ - Sortie diagnostic (contact sec Normalement Ouvert 11-14).

① - Resorte de clipsado en perfil de 35 mm.
② - Marcador con enganche.
③ - LED de estado de la tensión y de la corriente de salida **Uout - Iout**.
④ - Diagnóstico de salida (contacto seco Normalmente Abierto 11-14).

① - Mola de engate no perfilado de 35 mm.
② - Marca de engatar.
③ - LED de estado da tensão e da corrente de saída **Uout - Iout**.
④ - Resultado de Diagnóstico (contacto seco Normalmente Aberto 11-14).

U In	I out	ABL		U In	I out	ABL	
1 Ph ~ 100...120 V	3 A 5 A 10 A	8RPS24030 8RPS24050 8RPS24100		1 Ph ~ 200...500 V	3 A 5 A 10 A	8RPS24030 8RPS24050 8RPS24100	
1 Ph ~ 100...120 V	20 A	8RPM24200		1 Ph ~ 200...240 V	20 A	8RPM24200	
2 Ph ~ 200...500 V	3 A 5 A 10 A	8RPS24030 8RPS24050 8RPS24100		3 Ph ~ 380...500 V	20 A 40 A	8WPS24200 8WPS24400	

mm in	mm ² AWG	mm ² /AWG	mm/In	mm ² /AWG
10 0.39	Ø ≤ 4 mm ² Ø ≤ 12 AWG	1...4 / 16...12	4 / 0.16	0.2...2.5 / 24...14
17 0.67	Ø > 4 mm ² Ø > 12 AWG	1...4 / 16...12	4 / 0.16	0.2...2.5 / 24...14

Ø ≤ 4 mm ² Ø ≤ 12 AWG	Ø > 4 mm ² Ø > 12 AWG
Nm 0,6	2
lb-in 5.4	17.7

⚠ DANGER / DANGER / GEFAHR / PELIGRO / PERICOLO / PERIGO

HAZARD OF ELECTRIC SHOCK, EXPLOSION OR ARC FLASH

Disconnect all power before servicing equipment.

Failure to follow these instructions will result in death or serious injury.

RISQUE D'ELECTROCUTION, D'EXPLOSION OU D'ARC ELECTRIQUE

Coupez l'alimentation avant de travailler sur cet appareil.

Le non-respect de ces instructions provoquera la mort ou des blessures graves.

STROMSCHLAG-, EXPLOSIONS- ODER LICHTBOGENGEFAHR

Vor dem Arbeiten an dem Gerät dessen Stromversorgung abschalten.

Die Nichtbeachtung dieser Anweisungen führt zu Tod oder schwerer Körperverletzung.

RIESGO DE ELECTROCUCIÓN, EXPLOSIÓN O ARCO ELÉCTRICO

Desconecte toda alimentación antes de realizar el servicio.

Si no se siguen estas instrucciones provocará lesiones graves o incluso la muerte.

RISCHIO DI SCOSSA ELETTRICA, DI ESPLOSIONE O DI OFTALMIA DA FLASH

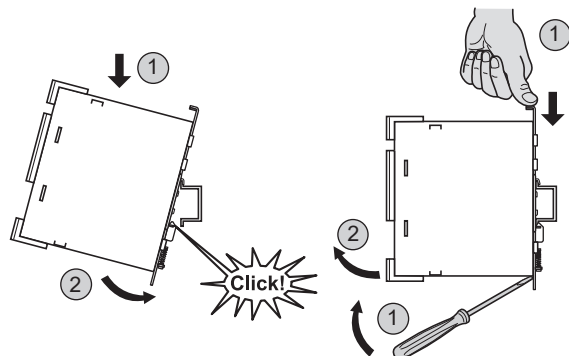
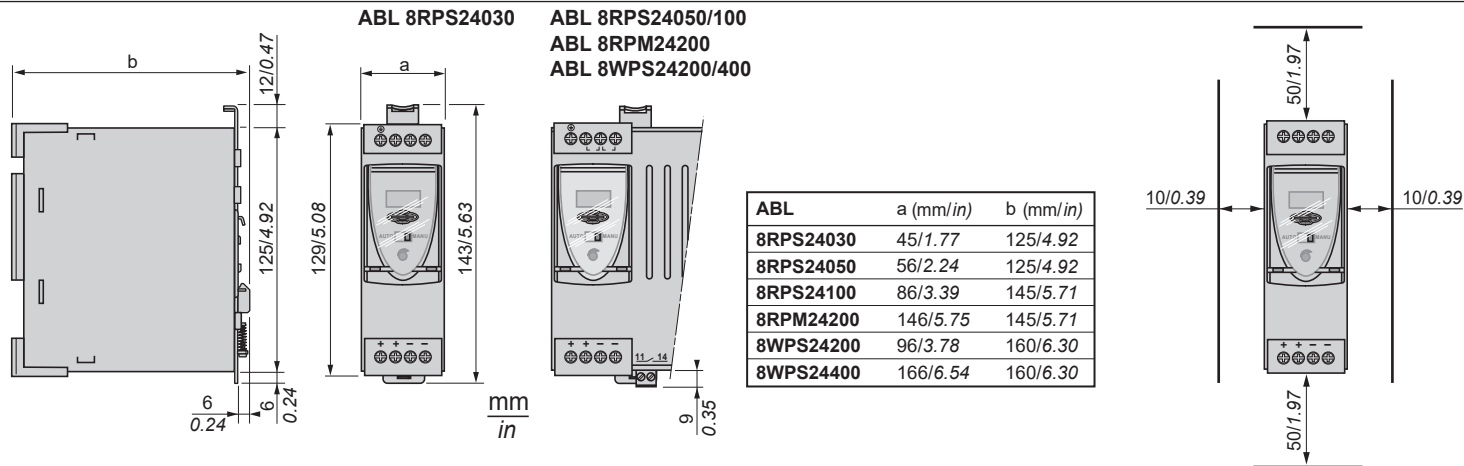
Scollegare l'apparecchio dalla presa di corrente prima di qualsiasi intervento.

Il mancato rispetto di queste istruzioni provocherà morte o gravi infortuni.

RISCO DE CHOQUE ELÉCTRICO, EXPLOÇÃO OU FAÍSCA

Desligue a alimentação antes de trabalhar neste aparelho.

A não observância destas instruções resultará em morte, ou ferimentos graves.

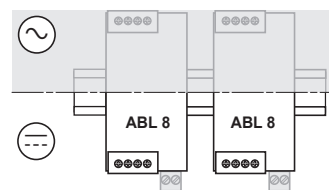
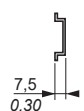
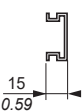
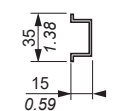


AM1 DE200
IEC/EN 60715

AM1 ED200

AM1 DP200

DZ5 MB200



U out		11 14
	$21,6 \text{ V} \leq U_{out}$	
	$7 \text{ V} \leq U_{out} < 21,6 \text{ V}$	
	$U_{out} < 7 \text{ V}$	

OFF

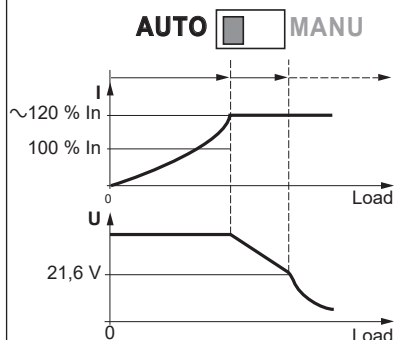
Green / Vert / Grün / Verde / Verde / Verde

Orange / Orange / Orange / Naranja / Arancione / Laranja

Red / Rouge / Rot / Rojo / Rosso / Vermelho

I out	
	$I_{out} \leq I_n$
	$I_{out} > I_n$
	Power deactivated after detection of overcurrent, overvoltage or overtemperature. (1) $0 \text{ V} / 0 \text{ A}$

(1) Alimentation arrêtée suite à une détection de surintensité, de surtension ou de surchauffe. Nach der Erkennung von Überstrom, Überspannung oder Übertemperatur wurde die Stromversorgung abgeschaltet. Alimentación desactivada tras detectar sobrecorriente, sobretensión o sobrecalentamiento. Alimentazione disattivata dopo il rilevamento di condizioni di sovracorrente, sovratensione o di surriscaldamento. Alimentação desactivada após a detecção de corrente, tensão ou temperatura excessivas.



AUTO

Constant current mode. Return to rated power supply operation once the source of overcurrent has been corrected.

AUTO

Mode courant constant. Retour au fonctionnement normal dès que l'origine de la surintensité a été corrigée.

AUTO

Konstantstrom-Modus. Rückkehr zum Nennbetrieb der Stromversorgung, sobald die Ursache des Überstroms behoben wurde.

AUTO

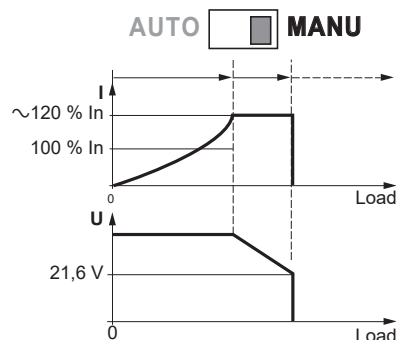
Modalidad de corriente constante. Vuelta al funcionamiento con fuente de alimentación nominal una vez corregido el origen de la sobrecorriente.

AUTO

Modalità di corrente costante. Ritornare al funzionamento con valori di alimentazione nominali una volta risolto il problema di sovracorrente.

AUTO

Modo de corrente constante. Voltar para o funcionamento com a alimentação de corrente nominal assim que a alimentação de corrente excessiva tiver sido corrigida.



MANU

Error retention. Following deactivation, remove power to the primary circuit and reapply power to the product again.

MANU

Mémorisation de surintensité. A la suite de l'arrêt de l'alimentation, couper l'alimentation au primaire et remettre le produit sous tension.

MANU

Fehlerspeicherung. Nach der Deaktivierung die Stromversorgung am Primärkreis abschalten und dem Produkt erneut Strom zuführen.

MANU

Error retenido. Tras completarse la desactivación, desconecte la alimentación del circuito primario y, a continuación, vuelva a aplicarla al producto.

MANU

Ritenzione errore. Dopo la disattivazione, rimuovere l'alimentazione al circuito principale, e rimettere di nuovo sotto tensione il prodotto.

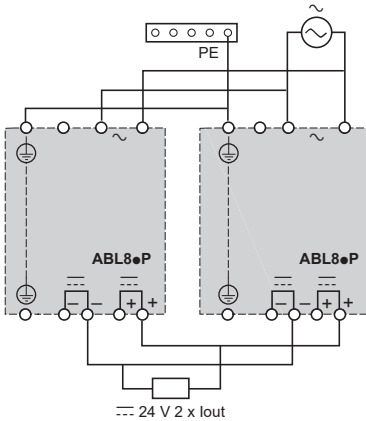
MANU

Retenção de erro. Após a desativação, remover a corrente para o circuito principal e voltar a aplicar corrente no produto.

⚠ 2 ABL 8●P max output //

SELV / TBTS

Outputs connected in parallel
Raccordement des sorties en parallèles
Anschluss der parallelgeschalteten Ausgänge
Conexión de las salidas en paralelo
Collegamento delle uscite in parallelo
Ligação das saídas em paralelas



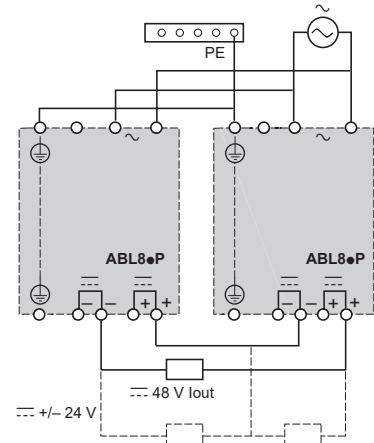
SELV / TBTS :

SELV: Safety Extra Low Voltage
TBTS: Très Basse Tension Sécurisée
Sicherheitskleinspannung
Muy baja tensión asegurada
Tensione di sicurezza molto bassa
Muito baixa tensão segura

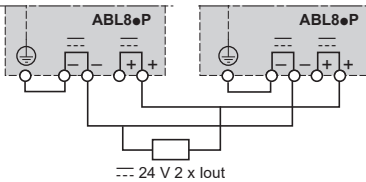
⚠ 2 ABL 8●P max output

SELV / TBTS

Series connection of the power supplies
Raccordement des alimentations en "série"
Anschluss der reihengeschalteten Stromversorgungen
conexión de las alimentaciones en "serie"
Collegamento delle alimentazioni in "serie"
Conexão das alimentações em "série"



PELV / TBTP



PELV / TBTP :

PELV: Protection Extra Low Voltage
TBTP: Très Basse Tension Protection
Schutzkleinspannung
Muy baja tensión protección
Tensione di protezione molto bassa
Muito baixa tensão protecção

Paralleling:

- Use maximum of 2 power supplies of the same reference.

Mise en parallèle:

- Utilisez 2 alimentations maximum et de même référence.

Parallelschaltung:

- max. 2 Stromversorgungen und nur bei gleichen Modellen.

Puesta en paralelo:

- 2 alimentaciones como máximo, con la misma referencia.

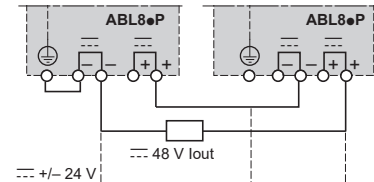
Messa in parallelo:

- 2 alimentazioni al massimo e delle stesse caratteristiche.

Ligação em paralelo:

- 2 alimentações no máximo e da mesma referência.

PELV / TBTP



WARNING / AVERTISSEMENT / WARNUNG / ADVERTENCIA / AVVERTENZA / AVISO

RISK OF MATERIAL DAMAGE AND HOT ENCLOSURE

- Allow the product sufficient time to cool before touching.
- Follow proper mounting instructions including torque values and the crimping lengths on wire terminations.
- Do not allow liquids or foreign objects to enter this product.

Failure to follow this instruction can result in death, serious injury, or equipment damage.

RISQUE DE DOMMAGE MATERIEL ET DE SURCHAUFFE DU BOITIER

- Laisser le produit refroidir avant de le toucher.
- Respecter les consignes de montage, et notamment les couples de serrage et les longueurs de sertissage sur les terminaisons de câble.
- Ne pas laisser pénétrer de liquide ni de corps étrangers à l'intérieur du produit.

Le non-respect de cette directive peut entraîner la mort, des lésions corporelles graves ou des dommages matériels.

GEFAHR VON MATERIALSCHÄDEN UND GEHÄUSEERHITZUNG

- Lassen Sie das Produkt lange genug abkühlen, bevor Sie es berühren.
- Beachten Sie die Montageanweisungen, insbesondere die Anziehdrehmomente und die Crimplängen an den Kabelenden.
- Führen Sie keine Flüssigkeiten oder Fremdkörper in das Produkt ein.

Die Nichtbeachtung dieser Anweisung kann den Tod, Körperverletzung oder Materialschäden zur Folge haben.

RIESGO DE DAÑOS MATERIALES Y DE SOBRECIENTAMIENTO DE LA UNIDAD

- Espere el tiempo necesario hasta que se enfríe el producto antes de tocarlo.
- Respetar las instrucciones de montaje, y en particular los pares de apretado y las longitudes de engaste en las terminaciones de los cables.
- No dejar que penetren líquidos o cuerpos extraños en el producto.

Si no se respetan estas precauciones pueden producirse graves lesiones, daños materiales o incluso la muerte.

RISCHIO DI DANNI MATERIALI E D'INVOLUCRO CALDO

- Attendere il raffreddamento del prodotto prima di toccarlo.
- Seguire le istruzioni di montaggio corrette che comprendono i valori di coppia e le lunghezze di crimpatura sulle terminazioni dei cavi.
- Non far entrare liquidi o oggetti estranei in questo apparecchio.

La mancata osservanza di questa precauzione può causare gravi rischi per l'incolumità personale o danni alle apparecchiature.

RISCO DE DANO MATERIAL E DE AQUECIMENTO

- Dar tempo suficiente para o produto arrefecer antes de lhe tocar.
- Siga devidamente as instruções de montagem, incluindo as forças de aperto e os comprimentos de enrolamento nos terminais de cabos.
- Não permita a entrada de líquidos e de objectos estranhos no produto.

A não observância destas precauções pode provocar a morte, ferimentos graves ou danos materiais.

Environment characteristics

Installation in a pollution degree 2 environment.
Maximum surrounding air temperature 50 °C (122 °F).
Minimum temperature rating of the conductor wires connected to the terminals :
75 °C (167 °F).

Caractéristiques d'environnement

Installation dans un environnement au niveau de pollution 2.
Température ambiante maximale 50 °C (122 °F).
Température nominale minimale des fils conducteurs raccordés aux bornes :
75 °C (167 °F).

Umgebungskenndaten

Installation in einer Umgebung mit Verschmutzungsgrad 2.
Maximale Umgebungstemperatur 50 °C (122 °F).
Die minimale Temperaturbemessung der Leitungsdrähte, die an die Klemmen angeschlossen sind, beträgt : 75 °C (167 °F).

Características ambientales

Instalación en un ambiente de contaminación grado 2.
Temperatura ambiente máxima 50 °C (122 °F).
Temperatura nominal minimal de los hilos conductores conectados a los terminales : 75 °C (167 °F).

Caratteristiche ambientali

Installazione in un ambiente con grado d'inquinamento II.
Massima temperatura ambiente circostante consentita 50 °C (122 °F).
Temperatura minima di esercizio dei conduttori collegati ai morsetti :
75 °C (167 °F).

Características ambientais

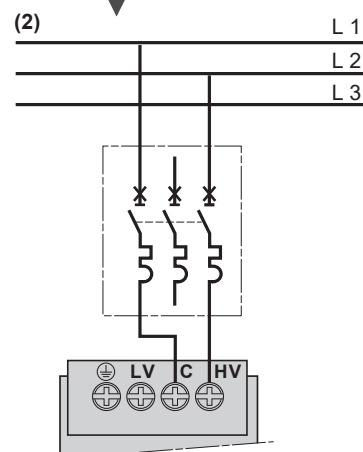
Instalação num ambiente de poluição nível 2.
Temperatura ambiental máxima 50 °C (122 °F).
Temperatura nominal mínima dos fios condutores conectados aos terminais :
75 °C (167 °F).

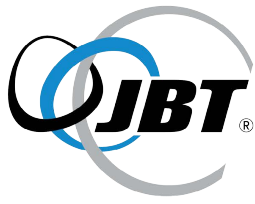
"Power supply modules, ABL8 series, UL certified Schneider Electric (E164867)"
Modules d'alimentation, série ABL8, certification UL Schneider Electric (E164867)
Versorgungsmodule, Baureihe ABL8, UL-zertifiziert, Schneider Electric (E164867)
"Módulos de alimentación, serie ABL8, con certificación UL de Schneider Electric (E164867)"
Moduli di alimentazione, serie ABL8, certificazione UL Schneider Electric (E164867)
Módulos de alimentação, série ABL8, certificação UL da Schneider Electric (E164867)

Selection of the protections on the power supply primary circuit
Choix des protections au primaire des alimentations
Wahl der Schutzart am Primärkreis der Versorgungen
Selección de las protecciones en el circuito primario de las alimentaciones
Scelta delle protezioni al primario delle alimentazioni
Seleção das proteções no circuito primário das alimentações
To be ordered separately / A commander séparément / Separat zu bestellen
Pedir por separado / Da comandare separatamente / Encomendar separadamente

ABL		~ 115 V						~ 230 V						~ 400 V					
		Outside of USA & Canada (IEC)			For USA and Canada (1)			Outside of USA & Canada (IEC)			For USA and Canada (1)			Outside of USA & Canada (IEC)			For USA and Canada (1)		
		gG/gL			Class CC rejection type			gG/gL			Class CC rejection type			gG/gL			Class CC rejection type		
8RPS24030	GB2/GV2	GB2 CD07	2 A	—	2 A	—	2 A	GB2 CD07	2 A	—	2 A	—	2 A	GV2 RT06 (2)	2 A	—	2 A	—	2 A
	C60N	2 A C curve	(8 x 32)	24443	(8 x 32)	24443	(8 x 32)	2 A C curve	(8 x 32)	24443	(8 x 32)	24443	(8 x 32)	—	2 A	(10,3 x 38,1)	2 A	(10,3 x 38,1)	2 A
8RPS24050	GB2/GV2	GB2 CD08	4 A	—	4 A	—	4 A	GB2 CD07	2 A	—	2 A	—	2 A	GV2 RT06 (2)	2 A	—	2 A	—	2 A
	C60N	3 A C curve	(8 x 32)	24444	(8 x 32)	24444	(8 x 32)	2 A C curve	(8 x 32)	24443	(8 x 32)	24443	(8 x 32)	—	2 A	(10,3 x 38,1)	2 A	(10,3 x 38,1)	2 A
8RPS24100	GB2/GV2	GB2 CD12	6 A	—	6 A	—	6 A	GB2 CD08	4 A	—	4 A	—	4 A	GV2 RT07 (2)	4 A	—	4 A	—	4 A
	C60N	6 A C curve	(8 x 32)	24447	(8 x 32)	24447	(8 x 32)	3 A C curve	(8 x 32)	24444	(8 x 32)	24444	(8 x 32)	—	4 A	(10,3 x 38,1)	4 A	(10,3 x 38,1)	4 A
8RPM24200	GB2/GV2	GB2 CD16	10 A	—	10 A	—	10 A	GB2 CD12	6 A	—	6 A	—	6 A						
	C60N	10 A C curve	(8 x 32)	24449	(8 x 32)	24449	(8 x 32)	6 A C curve	(8 x 32)	24447	(8 x 32)	24447	(8 x 32)						
8WPS24200	GB2/GV2													GV2 ME06	2 A		2 A		2 A
	C60N													—	(10,3 x 38,1)		(10,3 x 38,1)		(10,3 x 38,1)
8WPS24400	GB2/GV2													GV2 ME07	4 A		4 A		4 A
	C60N													—	(10,3 x 38,1)		(10,3 x 38,1)		(10,3 x 38,1)

(1) Conformance with UL508 and CSA 22.2 n°14





PRIME EQUIPMENT GROUP

REVISION PAGE

Revision	Description	Initials	Date